BAYESIAN STATISTICS B

Applications of Bayesian Statistics

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Aims Of This Course

1. Review of basic concepts in Bayesian statistics.
2. Learn basic principles of Markov chain Monte Carlo (MCMC) methods.
3. Study about standard Markov chain sampling algorithms such as Gibbs sampling algorithm, Metropolis-Hastings algorithm, and Hamiltonian Monte Carlo.
4. Study about the basic principle and real-world applications of advanced Bayesian statistical methods such as shrinkage priors, discrete choice models, quantile regression models, and hierarchical modeling.
5. Hands-on practice of Python and PyMC.

Reading List i

1. Introduction to Bayesian statistics
    - Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A. and Rubin, D.B. (2013). *Bayesian Data Analysis*, 3rd ed., Chapman & Hall/CRC.
      Download the book here.
    - Hoff, P.D. (2010). *A First Course in Bayesian Statistical Methods*, Springer.
      Download the book here (Keio account required).
    - Martin, O. (2024). Bayesian Analysis with Python: A practical guide to probabilistic modeling, 3rd ed., Packt Publishing.

Reading List ii

2. Advanced topics in Bayesian statistics
    - Chan, J., Koop, G., Poirier, D.J. and Tobias, J.L. (2020). *Bayesian Econometric Methods*, 2nd ed., Cambridge University Press.
    - Durbin, J. and Koopman, S.J. (2012). *Time Series Analysis by State Space Methods*, 2nd ed., Oxford University Press.
    - Prado, R. and West, M. (2021). *Time Series: Modeling, Computation, and Inference*, 2nd ed., Chapman & Hall/CRC.
    - Rossi, P.E., Allenby, G.E. and McCulloch, R. (2005). *Bayesian Statistics and Marketing*, Wiley.

Summary Of Bayesian Analysis

- θ — $m \times 1$ vector of unknown parameters
- D — data
- $p(x|\theta)$ — population distribution
- $p(\theta)$ — prior distribution
- $p(D|\theta)$ — likelihood
- $p(\theta|D)$ — posterior distribution

Bayes' Theorem

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{\int_{\mathbb{R}^m} p(D|\theta)p(\theta)d\theta}.$$

Difficulties In Bayesian Analysis

1. In Bayesian analysis, we are required to evaluate multiple integrals to proceed statistical inference.
2. For example, we need $\int_{\mathbb{R}^m} p(D|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$ to derive the exact expression of the posterior distribution, but it is not available except for a limited number of examples (e.g., natural conjugate prior).
3. It is the case for the following quantities:
   - Marginal distribution: $p(\theta_j|D) = \int_{\mathbb{R}^{m-1}} p(\boldsymbol{\theta}|D)d\boldsymbol{\theta}_{-j}$.
   - Mean: $\mathbf{E}_{\boldsymbol{\theta}}[\theta_j|D] = \int_{-\infty}^{\infty} \theta_j p(\theta_j|D)d\theta_j$.
   - Median: $\int_{-\infty}^{\mathrm{Median}_{\theta_j}} p(\theta_j|D)d\theta_j = \frac{1}{2}$.
   - Probability: $\mathbf{P}(a \leqq \theta_j \leqq b|D) = \int_a^b p(\theta_j|D)d\theta_j$.
   - Predictive dist.: $p(\tilde{x}|D) = \int_{\mathbb{R}^m} p(\tilde{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}|D)d\boldsymbol{\theta}$.

MAP Estimation

One exception is the posterior mode:

$$\text{Mode}_{\boldsymbol{\theta}} = \mathop{\mathbf{arg\,max}}_{\boldsymbol{\theta} \in \mathbb{R}^m} p(\boldsymbol{\theta}|D).$$

Note that the posterior mode is equivalent to

$$\text{MAP}_{\boldsymbol{\theta}} = \mathop{\mathbf{arg\,max}}_{\boldsymbol{\theta} \in \mathbb{R}^m} p(D|\boldsymbol{\theta})p(\boldsymbol{\theta}),$$

which does not depend on the normalizing constant. Since the exact expressions of the prior distribution and the likelihood are known in many applications, we can solve the above optimization problem without evaluating the normalizing constant.

The posterior mode is often called MAP (maximum a posteriori) estimate.

Monte Carlo Methods i

Monte Carlo methods are widely applied numerical techniques to evaluate integrals. Suppose we need to evaluate the following expectation:

$$\mathbf{E}_{\theta}[h(\theta)] = \int_{\mathbb{R}^m} h(\theta)p(\theta|D)d\theta.$$

Examples:

- Mean $\mathbf{E}_{\theta}[\theta_j|D]$: $h(\theta) = \theta_j$.
- Probability $\mathbf{P}(a \leqq \theta_j \leqq b|D)$: $h(\theta) = \mathbb{1}_{[a,b]}(\theta_j)$.
- Predictive distribution $p(\tilde{x}|D)$: $h(\theta) = p(\tilde{x}|\theta)$.

Monte Carlo Methods ii

Suppose we have a set of pseudo-random numbers generated from the posterior distribution $p(\theta|D)$, $\{\theta^{(1)}, \ldots, \theta^{(T)}\}$, which is called the Monte Carlo sample, and define

$$\hat{h} = \frac{1}{T}\sum_{t=1}^{T} h(\theta^{(t)}).$$

If the law of large numbers holds,

$$\hat{h} \rightsquigarrow \mathbf{E}_{\theta}[h(\theta)] \quad \text{as} \quad T \to \infty,$$

where "$\rightsquigarrow$" means the convergence in probability.

Monte Carlo Methods iii

Therefore, when T is sufficiently large, $\mathbf{E}_{\theta}[h(\theta)]$ can be well approximated with $\hat{h}$. This is the basic idea behind the Monte Carlo integration method.

Monte Carlo approximations of the posterior statistics are given as follows:

- Mean: $\hat{\mathbf{E}}_{\theta}[\theta_j|D] = \frac{1}{T}\sum_{t=1}^{T}\theta_j^{(t)}$.
- Probability: $\widehat{\mathbf{P}}(a \leqq \theta_j \leqq b|D) = \frac{1}{T}\sum_{t=1}^{T}\mathbb{1}_{[a,b]}(\theta_j^{(t)})$.
- Predictive distribution: $\hat{p}(\tilde{x}|D) = \frac{1}{T}\sum_{t=1}^{T}p(\tilde{x}|\theta^{(t)})$.

Note that the Monte Carlo method merely gives us an approximated value of the true posterior statistic (the true one is obtained when $T \to \infty$). So it is contaminated with numerical errors. These errors are measured by

Numerical errors in Monte Carlo approximation

$$\mathbf{SE}[\hat{h}] = \sqrt{\frac{1}{T(T-1)} \sum_{t=1}^{T} \left(h(\theta^{(t)}) - \hat{h}\right)^2}.$$

Kernel Density Estimation

The marginal posterior p.d.f. $p(\theta_j|D)$ can be evaluated with a kernel density estimation method:

$$\hat{p}(\theta_j) = \frac{1}{Tw}\sum_{t=1}^{T} K\left(\frac{\theta_j - \theta_j^{(t)}}{w}\right),$$

where $K(\cdot)$ is a kernel function. In practice, the p.d.f. of the standard normal distribution:

$$\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}},$$

is often used as $K(\cdot)$, which is called the Gaussian kernel. w is the bandwidth and determines the smoothness of the evaluated density function.

Monte Carlo Approximation Of Quantiles i

Quantile

The $100q\%$ quantile $\theta_j^{[q]}$ is defined as

$$\mathbf{P}\left(\theta_j \leqq \theta_j^{[q]} \middle| D\right) = q, \quad 0 < q < 1.$$

We can approximate the quantile $\theta_j^{[q]}$ with

Sample Quantile

$$\hat{\theta}_j^{[q]} = \max_{1 \leqq t \leqq T} \theta_j^{(t)} \quad \text{s.t.} \quad \frac{1}{T} \sum_{s=1}^{T} \mathbb{1}_{\left(-\infty,\, \theta_j^{(t)}\right)}\left(\theta_j^{(s)}\right) \leqq q,$$

Monte Carlo Approximation Of Quantiles ii

For example, when $T = 10,000$, $\hat{\theta}_j^{[0.05]}$ is the 500th smallest value in the Monte Carlo sample.

- Median: $\hat{\theta}_j^{[0.5]}$.
- $100(1-q)\%$ credible interval: $\hat{\theta}_j^{[\frac{q}{2}]} \leqq \theta_j \leqq \hat{\theta}_j^{[1-\frac{q}{2}]}$.
- $100(1-q)\%$ HPDI [Chen and Shao (1999)]:

$$\hat{\theta}_j^{\left[\frac{t^*}{T}\right]} \leqq \theta_j \leqq \hat{\theta}_j^{\left[1-q+\frac{t^*}{T}\right]}, \quad 1 \leqq t \leqq qT,$$

where

$$t^* = \arg\min_{1\leqq t\leqq qT} \left|\hat{\theta}_j^{[\frac{t}{T}]} - \hat{\theta}_j^{[1-q+\frac{t}{T}]}\right|.$$

Markov Chain Monte Carlo (MCMC)

- So far we suppose we have the Monte Carlo sample generated from the posterior distribution.
- However it is not obvious how to generate random numbers from an unknown distribution.
- For this purpose, we apply Markov chain sampling methods (e.g., Gibbs sampler, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo method).
- Markov chain sampling + Monte Carlo integration = Markov chain Monte Carlo (MCMC)
- MCMC is indispensable for Bayesian statistics.
- To understand MCMC, we need to know a Markov chain.

Markov Chain i

Since

$$\mathbf{P}(A) = \mathbf{P}(A|B)\mathbf{P}(B) + \mathbf{P}(A|B^c)\mathbf{P}(B^c),$$
$$\mathbf{P}(A^c) = \mathbf{P}(A^c|B)\mathbf{P}(B) + \mathbf{P}(A^c|B^c)\mathbf{P}(B^c).$$

we have

$$\begin{bmatrix} \mathbf{P}(A) \\ \mathbf{P}(A^c) \end{bmatrix} = \begin{bmatrix} \mathbf{P}(A|B) & \mathbf{P}(A|B^c) \\ \mathbf{P}(A^c|B) & \mathbf{P}(A^c|B^c) \end{bmatrix} \begin{bmatrix} \mathbf{P}(B) \\ \mathbf{P}(B^c) \end{bmatrix}.$$

Markov Chain ii

Let A_t denote the event that A occur at time t and we replace A and B with A_{t+1} and A_t respectively. Then

$$\begin{bmatrix} \mathbf{P}(A_{t+1}) \\ \mathbf{P}(A_{t+1}^c) \end{bmatrix} = \begin{bmatrix} \mathbf{P}(A_{t+1}|A_t) & \mathbf{P}(A_{t+1}|A_t^c) \\ \mathbf{P}(A_{t+1}^c|A_t) & \mathbf{P}(A_{t+1}^c|A_t^c) \end{bmatrix} \begin{bmatrix} \mathbf{P}(A_t) \\ \mathbf{P}(A_t^c) \end{bmatrix}.$$

This is called a Markov chain. In the context of the Markov chain, each event is often referred to as a state and the conditional probability $\mathbf{P}(A_{t+1}|A_t^c)$ is interpreted as the probability that the chain moves from state A^c at t to state A at $t+1$.

Markov Chain iii

The matrix

$$\begin{bmatrix} \mathbf{P}(A_{t+1}|A_t) & \mathbf{P}(A_{t+1}|A_t^c) \\ \mathbf{P}(A_{t+1}^c|A_t) & \mathbf{P}(A_{t+1}^c|A_t^c) \end{bmatrix}.$$

is called the transition matrix. When the transition matrix does not change over time, the Markov chain is time-homogeneous. The vector of probabilities at time $t = 0$ is called the initial probability vector.

In economics, finance and other fields, the Markov chain is widely used for modeling a time-varying probability.

Let $p_{1t} = \mathbf{P}(A_t)$, $p_{2t} = \mathbf{P}(A_t^c)$, and let π_{ij} denote the (i, j) element of the time-homogeneous transition matrix $(i, j = 1, 2)$. Then the Markov chain is rewritten as

$$\begin{bmatrix} p_{1,t+1} \\ p_{2,t+1} \end{bmatrix} = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} \begin{bmatrix} p_{1,t} \\ p_{2,t} \end{bmatrix}.$$

In general, the Markov chain with k states is given by

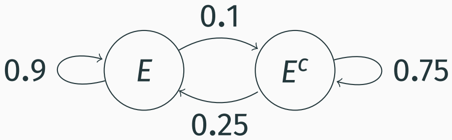
$$\underbrace{\begin{bmatrix} p_{1,t+1} \\ \vdots \\ p_{k,t+1} \end{bmatrix}}_{p_{t+1}} = \underbrace{\begin{bmatrix} \pi_{11} & \cdots & \pi_{1k} \\ \vdots & \ddots & \vdots \\ \pi_{k1} & \cdots & \pi_{kk} \end{bmatrix}}_{\mathbf{\Pi}} \underbrace{\begin{bmatrix} p_{1,t} \\ \vdots \\ p_{k,t} \end{bmatrix}}_{p_t}.$$

Example: Business Cycle

Suppose that the business cycle of a country is determined by the following transition matrix:

Table 1: Transition Matrix $\mathbf{\Pi}$

	Previous Quarter	
Current Quarter	Expansion (E)	Recession (E^c)
Expansion (E)	0.9	0.25
Recession (E^c)	0.1	0.75



Properties i

1. Regularity
   A Markov chain is regular if there exists a finite $t \geqq 1$ such that $\mathbf{P}(A_t = A_i | A_0 = A_j) > 0$ for any pair of (A_i, A_j). In other words, states in a regular Markov chain must be accessible from anywhere.
2. Aperiodicity
   Consider any $t \geqq 1$ satisfies $\mathbf{P}(A_t = A_i | A_0 = A_i) > 0$. The greatest common divisor of such t is called the period of A_i. A Markov chain is aperiodic if the period is one for any state.

Properties ii

3. Recurrence
   Define $\tau_i = \inf\{t \geqq 1 : A_t = A_i\}$. A_i is recurrent if $\mathbf{P}(\tau_i < \infty | A_0 = A_i) = 1$. A Markov chain is recurrent if all states are recurrent.
4. Time-homogeneity
   A Markov chain is time-homogeneous if $\mathbf{P}(A_{t+1} = A_i | A_t = A_j) = \mathbf{P}(A_t = A_i | A_{t-1} = A_j)$ for any pair of (A_i, A_j) and $t \geqq 1$.
5. $p_{t+k} = \mathbf{\Pi}^k p_t \ (k = 1, 2, \dots)$ when $\mathbf{\Pi}$ is constant. In particular, $p_t = \mathbf{\Pi}^t p_0$. Thus p_t is determined by t, p_0 and $\mathbf{\Pi}$ if the Markov chain is time-homogeneous.

Example: Transition Matrices

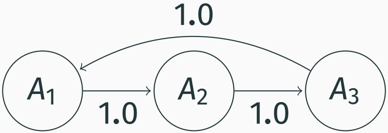
$$\underbrace{\begin{bmatrix} 0.9 & 0.1 & 0 & 0 \\ 0.1 & 0.9 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0.8 & 0.2 \end{bmatrix}}_{\mathbf{\Pi}_1}, \quad \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\mathbf{\Pi}_2}, \quad \underbrace{\begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.1 & 0.8 & 0 \\ 0 & 0.1 & 1 \end{bmatrix}}_{\mathbf{\Pi}_3}.$$

- $\mathbf{\Pi}_1$ is not regular. State 1 and 2 are not accessible from State 3 and 4, and vice versa.
- $\mathbf{\Pi}_2$ is periodic. The period is 3.
- State 1 and 2 in $\mathbf{\Pi}_3$ are not recurrent. State 3 is absorbing.

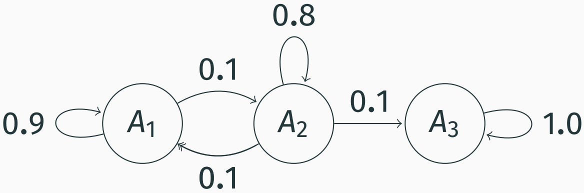
Π_1:


0.9
0.2
0.1
0.8
0.9
A_1
A_2
A_3
A_4
0.2
0.1
0.8

Π_2:


1.0
A_1
A_2
A_3
1.0
1.0

Π_3:


0.8
0.1
0.9
A_1
A_2
0.1
A_3
1.0
0.1

Stationary Distribution

1. The probability vector $\boldsymbol{p}^*$ such that

$$p^* = \mathbf{\Pi} p^*,$$

   is called the stationary probability vector or stationary distribution.

2. When the number of states is two, the stationary distribution is given by

$$p_1^* = \frac{1 - \pi_{22}}{2 - \pi_{11} - \pi_{22}}, \quad p_2^* = \frac{1 - \pi_{11}}{2 - \pi_{11} - \pi_{22}}.$$

3. Ergodicity
   If a time-homogeneous Markov chain is recurrent and aperiodic, $\boldsymbol{p}^* = \lim_{t\to\infty} \mathbf{\Pi}^t \boldsymbol{p}_0$ for any $\boldsymbol{p}_0 > 0$.

Example: Business Cycle

For the transition matrix

$$\mathbf{\Pi} = \begin{bmatrix} 0.9 & 0.25 \\ 0.1 & 0.75 \end{bmatrix},$$

the stationary distribution is given by

$$\mathbf{P}(E) = \frac{1 - 0.75}{2 - 0.9 - 0.75} = \frac{5}{7}, \quad \mathbf{P}(E^c) = 1 - \mathbf{P}(E) = \frac{2}{7}.$$

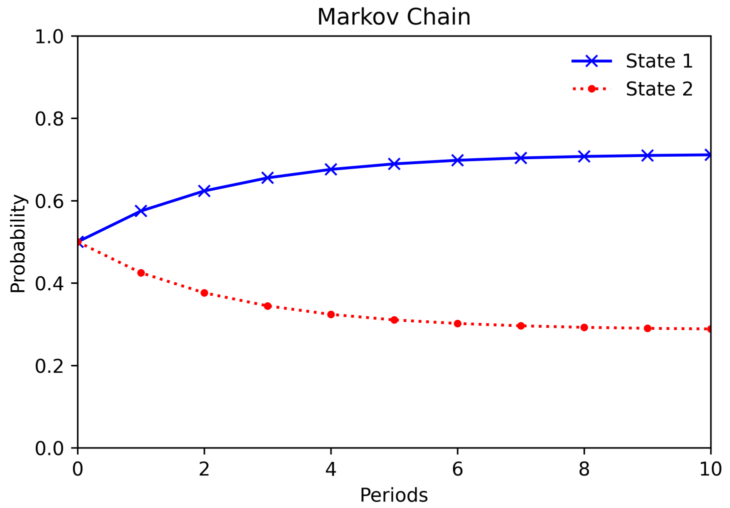


Figure 1: Convergence to the Stationary Distribution

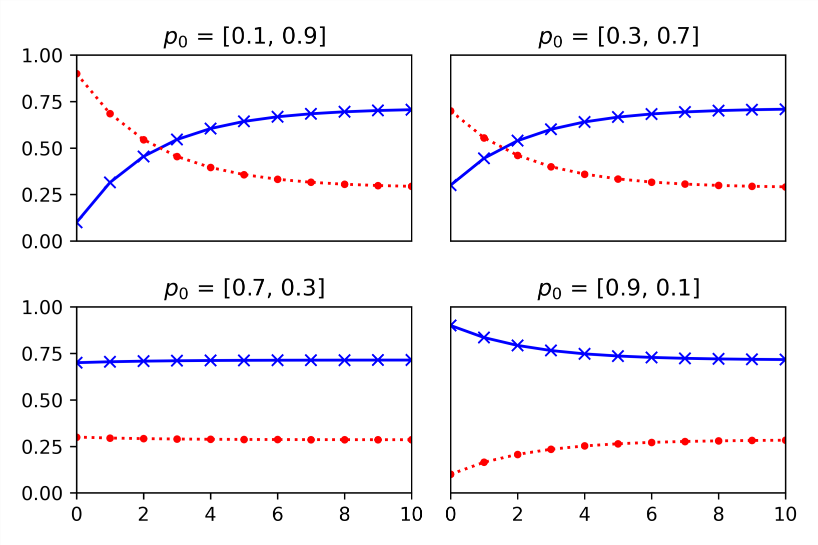


Figure 2: Influence of the Initial Probability Vector

Markov Chain Of Continuous Random Variables

Consider a sequence of continuous random variables $\{X_t\}_{t=0}^{\infty}$. X_t takes a real value in $\mathcal{X} \subseteqq \mathbb{R}$.

$\{X_t\}_{t=0}^{\infty}$ is called a Markov chain if, given $\{x_s\}_{s=0}^{t-1}$, the conditional probability that X_t takes a real value in $A \subseteqq \mathcal{X}$ is expressed as

$$\begin{aligned}\mathbf{P}(X_t \in A | X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) \\ = \mathbf{P}(X_t \in A | X_{t-1} = x_{t-1}).\end{aligned}$$

Properties i

- Time-homogeneity
  For any $A \subseteqq \mathcal{X}$, $x \in \mathcal{X}$, $t \geqq 0$,
  $\mathbf{P}(X_{t+1} \in A | X_t = x) = \mathbf{P}(X_t \in A | X_{t-1} = x)$.
- Regularity
  Suppose $\int_A f(x)dx > 0$ for any $A \subseteqq \mathcal{X}$. A Markov chain $\{X_t\}_{t=0}^{\infty}$ is regular with respect to f or f-regular if there exists a finite $t \geqq 1$ such that
  $\mathbf{P}(X_t \in A | X_0 = x) > 0$ for any $x \in \mathcal{X}$.

Properties ii

- Aperiodicity
  Consider any $t \geqq 1$ satisfies $\mathbf{P}(X_t \in A | X_0 = x) > 0$ for any $x \in A \subseteqq \mathcal{X}$. The greatest common divisor of such t is called the period. If the period is one for any $A \subseteqq \mathcal{X}$, a Markov chain $\{X_t\}_{t=0}^{\infty}$ is aperiodic.
- Recurrence
  Define $\tau_A = \inf\{t > 0 : X_t \in A\}$. A is recurrent if $\mathbf{P}(\tau_A < \infty | X_0 = x) = 1$ for any $x \in A$. $\{X_t\}_{t=0}^{\infty}$ is recurrent with respect to f if A is recurrent for a f-regular Markov chain $\{X_t\}_{t=0}^{\infty}$ and $\int_A f(x)dx > 0$.

Transition Kernel i

The conditional p.d.f. of X_t given $X_0, \ldots, X_{t-1}$ is

$$f_t(x_t | x_0, \ldots x_{t-1}) = f(x_t | x_{t-1}),$$

which is called the transition kernel, and the right-hand side is often expresses as $K(x_{t-1}, x_t)$.

Suppose $f_0(x_0)$ is the p.d.f. of the initial distribution. Then the joint p.d.f. of $\{X_s\}_{s=0}^t$ is

$$\begin{aligned} f(x_0, \ldots, x_t) &= f_0(x_0) f_1(x_1 | x_0) f_2(x_2 | x_0, x_1) \times \cdots \\ &\times f_t(x_t | x_0, \ldots, x_{t-1}) = f_0(x_0) \prod_{s=1}^{t} K(x_{s-1}, x_s). \end{aligned}$$

Transition Kernel ii

Note that

$$
\begin{aligned}
f_t(x_t) &= \int_{\mathcal{X}} f(x_{t-1}, x_t) dx_{t-1} \\
&= \int_{\mathcal{X}} f(x_t | x_{t-1}) f_{t-1}(x_{t-1}) dx_{t-1} \\
&= \int_{\mathcal{X}} f_{t-1}(x_{t-1}) K(x_{t-1}, x_t) dx_{t-1}.
\end{aligned}
$$

Define

$$
f_t = f_{t-1} \circ K = \int_{\mathcal{X}} f_{t-1}(x_{t-1}) K(x_{t-1}, x_t) dx_{t-1}.
$$

Transition Kernel iii

Then

$$
\begin{aligned}
f_t &= f_{t-1} \circ K = \left\{ \int_{\mathcal{X}} f_{t-2}(x_{t-2}) K(x_{t-2}, x_{t-1}) dx_{t-2} \right\} \circ K \\
&= \int_{\mathcal{X}} \left\{ \int_{\mathcal{X}} f_{t-2}(x_{t-2}) K(x_{t-2}, x_{t-1}) dx_{t-2} \right\} K(x_{t-1}, x_t) dx_{t-1} \\
&= \int_{\mathcal{X}} f_{t-2}(x_{t-2}) \underbrace{\int_{\mathcal{X}} K(x_{t-2}, x_{t-1}) K(x_{t-1}, x_t) dx_{t-1}}_{K \circ K} dx_{t-2} \\
&= f_{t-2} \circ (K \circ K) = f_{t-2} \circ K^2 \quad \cdots \quad = f_0 \circ K^t, \\
K^t &= \int_{\mathcal{X}} \cdots \int_{\mathcal{X}} K(x_0, x_1) \cdots K(x_{t-1}, x_t) dx_1 \ldots dx_{t-1}.
\end{aligned}
$$

Example: AR(1) Process i

AR(1) process

$$X_t = \phi X_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathbf{Normal}(0, \sigma^2), \quad |\phi| < 1, \quad \phi \neq 0.$$

This a Markov chain with the transition kernel:

$$f(x_t | x_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_t - \phi x_{t-1})^2}{2\sigma^2}\right],$$

which is the p.d.f. of the normal distribution $\mathbf{Normal}(\phi x_{t-1}, \sigma^2)$.

Example: AR(1) Process ii

The joint p.d.f. of $\{X_s\}_{s=0}^{t}$ is

$$
\begin{aligned}
f(x_0, \ldots, x_t) &= f_0(x_0) \prod_{s=1}^{t} f(x_s | x_{s-1}) \\
&= f_0(x_0) \prod_{s=1}^{t} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_s - \phi x_{s-1})^2}{2\sigma^2} \right].
\end{aligned}
$$

Invariant Distribution

The invariant distribution (density) $\bar{f}$ of a Markov chain with kernel K is

$$\bar{f}(\tilde{x}) = \int_{\mathcal{X}} \bar{f}(x)K(x,\tilde{x})dx \quad \text{or} \quad \bar{f} = \bar{f} \circ K.$$

If a Markov chain is recurrent and aperiodic,

Ergodicity: $$\lim_{t\to\infty} \sup_{A\subseteq\mathcal{X}} \left| \int_A (f_t(x) - \bar{f}(x))dx \right| = 0,$$

LLN: $$\frac{1}{T}\sum_{t=1}^{T} h(X_t) \rightsquigarrow \int_{\mathcal{X}} h(x)\bar{f}(x)dx.$$

Detailed Balance Condition (DBC)

$$\bar{f}(x)K(x,\tilde{x}) = \bar{f}(\tilde{x})K(\tilde{x},x), \quad {}^{\forall}x,\tilde{x} \in \mathcal{X}.$$

Proof Of DBC

By integrating both sides of DBC with respect to x, we have

$$\begin{aligned}\int_{\mathcal{X}} \bar{f}(x)K(x,\tilde{x})dx &= \int_{\mathcal{X}} \bar{f}(\tilde{x})K(\tilde{x},x)dx \\ &= \bar{f}(\tilde{x}) \int_{\mathcal{X}} K(\tilde{x},x)dx \\ &= \bar{f}(\tilde{x}),\end{aligned}$$

that is,

$$\bar{f}(\tilde{x}) = \int_{\mathcal{X}} \bar{f}(x)K(x,\tilde{x})dx,$$

holds. This means that $\bar{f}$ is the invariant density of K. ■

Example: AR(1) Process i

The invariant distribution of an AR(1) process is

$$\mathbf{Normal}\left(0, \frac{\sigma^2}{1-\phi^2}\right),$$

with the p.d.f.:

$$\bar{f}(x) = \sqrt{\frac{1-\phi^2}{2\pi\sigma^2}} \exp\left[-\frac{(1-\phi^2)x^2}{2\sigma^2}\right].$$

It suffices to show that the above p.d.f. satisfies DBC.

Example: AR(1) Process ii

$$
\begin{aligned}
\bar{f}(x)K(x,\tilde{x}) &= \frac{\sqrt{1-\phi^2}}{2\pi\sigma^2}\,\mathbf{exp}\left[-\frac{(1-\phi^2)x^2+(\tilde{x}-\phi x)^2}{2\sigma^2}\right] \\
&= \frac{\sqrt{1-\phi^2}}{2\pi\sigma^2}\,\mathbf{exp}\left[-\frac{x^2-\phi^2x^2+\tilde{x}^2-2\phi x\tilde{x}+\phi^2x^2}{2\sigma^2}\right] \\
&= \frac{\sqrt{1-\phi^2}}{2\pi\sigma^2}\,\mathbf{exp}\left[-\frac{\tilde{x}^2-\phi^2\tilde{x}^2+x^2-2\phi x\tilde{x}+\phi^2\tilde{x}^2}{2\sigma^2}\right] \\
&= \frac{\sqrt{1-\phi^2}}{2\pi\sigma^2}\,\mathbf{exp}\left[-\frac{(1-\phi^2)\tilde{x}^2+(x-\phi\tilde{x})^2}{2\sigma^2}\right] \\
&= \bar{f}(\tilde{x})K(\tilde{x},x).
\end{aligned}
$$

Example: AR(1) Process iii

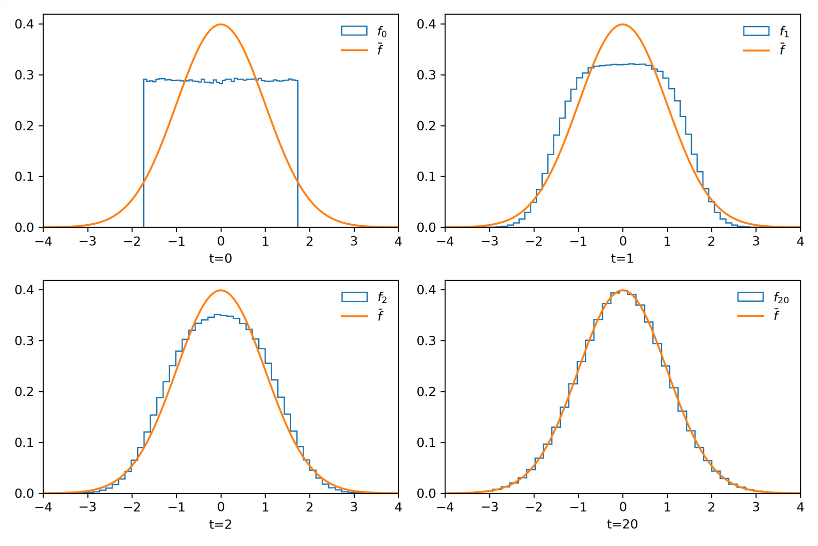
As a numerical illustration, we simulate the following AR(1) process:

$$
\begin{aligned}
x_0 &\sim \mathbf{Uniform}(-\sqrt{3}, \sqrt{3}), \\
x_t &= 0.9 x_{t-1} + \epsilon_t,\ \epsilon_t \sim \mathbf{Normal}(0, 0.19),\ t = 1, 2, \ldots
\end{aligned}
$$

The invariant distribution is the standard normal distribution $\mathbf{Normal}(0, 1)$ since

$$
\frac{\sigma^2}{1-\phi^2} = \frac{0.19}{1-0.9^2} = 1.
$$

Example: AR(1) Process iv



Random Number Generation From A Markov Chain

Markov chain sampling

Step 1. Set $t = 1$ and $\tilde{x}_0 \leftarrow f_0(x_0)$.

Step 2. $\tilde{x}_t \leftarrow K(\tilde{x}_{t-1}, x_t)$.

Step 3. Increase t by 1 and go to **Step 2**.

- Suppose we can generate $\tilde{x}_t$ from a recurrent and aperiodic Markov chain with K and $\bar{f}$.
- After we repeat Step 1–3 sufficiently many times (say M), the distribution of $\tilde{x}_t$ will be very close to $\bar{f}$.
- $\mathbb{E}[h(X)] = \int_{\mathcal{X}} h(x)\bar{f}(x)dx$ can be approximated by $\frac{1}{N}\sum_{t=M+1}^{M+N} h(\tilde{x}_t)$ due to ergodicity of the Markov chain.
- The first M runs of the Markov chain sampling is called the burnin.

Convergence Diagnostics

Unfortunately Markov chain sampling does not always work, even though the distribution of generated random numbers must converge to the posterior distribution in theory.

To make it work properly, we need careful tuning and monitoring. PyMC can automatically tune the sampling algorithm to some extent. So we discuss how to check the convergence of generated chains.

- n — the number of draws
- m — the number of chains
- θ_{ij} — draw i in chain j $(i = 1, \ldots, n,\ j = 1, \ldots, m)$
- $\{\theta_1, \ldots, \theta_T\}$ — single-chain Monte Carlo sample

Numerical Errors For Correlated Samples

The numerical error of the sample mean is

$$\mathbf{SE}[\bar{\theta}] = \sqrt{\frac{1}{T(T-1)} \sum_{t=1}^{T} (\theta_t - \bar{\theta})^2}, \quad \bar{\theta} = \frac{1}{T} \sum_{t=1}^{T} \theta_t.$$

In practice, we observe strong positive autocorrelations in the Mote Carlo sample, which makes the above $\mathbf{SE}[\bar{\theta}]$ underestimate the true numerical error. Instead we use

$$\mathbf{SE}_S[\bar{\theta}] = \sqrt{\frac{1}{T}\left(\hat{\gamma}_0 + 2\sum_{s=1}^{S} w\left(\frac{s}{S+1}\right)\hat{\gamma}_s\right)},$$

where $\hat{\gamma}_s = \frac{1}{T}\sum_{t=s+1}^{T}(\theta_t - \bar{\theta})(\theta_{t-s} - \bar{\theta})$ and $w(\cdot)$ is called a lag window.

Lag Windows

- Rectangular lag window (used in PyMC by default)

$$w(x) = 1, \quad (|x| \leqq 1).$$

- Triangular (Bartlett) lag window

$$w(x) = 1 - |x|, \quad (|x| \leqq 1).$$

- Parzen lag window

$$w(x) = \begin{cases} 1 - 6x^2 + 6|x|^3, & \left(|x| < \frac{1}{2}\right); \\ 2(1 - |x|)^3, & \left(\frac{1}{2} \leqq x \leqq 1\right); \\ 0, & (\text{otherwise}). \end{cases}$$

Numerical Inefficiency

Effective sample size

$$\hat{T}_e = \frac{T}{1 + 2\sum_{s=1}^{S} w\left(\frac{s}{S+1}\right)\hat{\rho}_s}, \quad \hat{\rho}_s = \frac{\hat{\gamma}_s}{\hat{\gamma}_0}$$

If T is large enough,

$$\begin{aligned}\mathbf{SE}_S[\bar{\theta}] &= \sqrt{\frac{\hat{\gamma}_0}{T}\left(1 + 2\sum_{s=1}^{S} w\left(\frac{s}{S+1}\right)\hat{\rho}_s\right)} \\ &\approx \mathbf{SE}[\bar{\theta}]\sqrt{1 + 2\sum_{s=1}^{S} w\left(\frac{s}{S+1}\right)\hat{\rho}_s} = \mathbf{SE}[\bar{\theta}]\sqrt{\frac{T}{\hat{T}_e}}.\end{aligned}$$

Thus $\mathbf{SE}_S[\bar{\theta}]$ will be inflated if $\hat{T}_e$ is small.

Gelman-Rubin Convergence Diagnostic

Gelman-Rubin convergence diagnostic $\hat{R}$

$$\hat{R} = \sqrt{\frac{\hat{V}}{\hat{W}}}, \quad \hat{V} = \frac{n-1}{n}\hat{W} + \frac{1}{n}\hat{B},$$

$$\hat{B} = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_{\cdot j} - \bar{\theta})^2, \quad \bar{\theta}_{\cdot j} = \frac{1}{n}\sum_{i=1}^{n}\theta_{ij},$$

$$\hat{W} = \frac{1}{m}\sum_{j=1}^{m}s_j^2, \quad s_j^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\theta_{ij} - \bar{\theta}_{\cdot j})^2.$$

If the mean of each chain is equal to each other, $\frac{1}{n}\hat{B}$ will be close to zero when n is sufficiently large. Therefore $\hat{R}$ must be close to one; otherwise, the convergence is doubtful.

Gibbs Sampler

Suppose a bivariate distribution $f(\theta_1, \theta_2)$. We cannot generate random numbers directly from it, but it is possible to generate them from both $f(\theta_1|\theta_2)$ and $f(\theta_2|\theta_1)$.

Then the Gibbs sampler is defined as follows.

Gibbs sampler for a bivariate distribution

Step 0. Set the initial states $(\theta_1^{(0)}, \theta_2^{(0)})$ and $r = 1$.

Step 1. $\theta_1^{(r)} \leftarrow f(\theta_1|\theta_2^{(r-1)})$.

Step 2. $\theta_2^{(r)} \leftarrow f(\theta_2|\theta_1^{(r)})$.

Step 3. $r \leftarrow r + 1$.

Transition Kernel

The transition kernel of the Gibbs sampler is

$$K(\theta^{(r-1)}, \theta^{(r)}) = f(\theta_2^{(r)} | \theta_1^{(r)}) f(\theta_1^{(r)} | \theta_2^{(r-1)}).$$

Proof: Let $\theta = \theta^{(r-1)}$ and $\tilde{\theta} = \theta^{(r)}$.

$$\begin{aligned}
\int f(\theta) K(\theta, \tilde{\theta}) d\theta &= \int \int f(\tilde{\theta}_2 | \tilde{\theta}_1) f(\tilde{\theta}_1 | \theta_2) f(\theta_1, \theta_2) d\theta_1 d\theta_2 \\
&= f(\tilde{\theta}_2 | \tilde{\theta}_1) \int f(\tilde{\theta}_1 | \theta_2) \left\{ \int f(\theta_1, \theta_2) d\theta_1 \right\} d\theta_2 \\
&= f(\tilde{\theta}_2 | \tilde{\theta}_1) \int f(\tilde{\theta}_1 | \theta_2) f(\theta_2) d\theta_2 \\
&= f(\tilde{\theta}_2 | \tilde{\theta}_1) f(\tilde{\theta}_1) = f(\tilde{\theta}_1, \tilde{\theta}_2) = f(\tilde{\theta}).
\end{aligned}$$

■

Remarks i

1. The invariant distribution of the Gibbs sampler is *invariant* to the order of sampling. If we switch the order as

Alternative Gibbs sampler

Step 0. Set the initial states $(\theta_1^{(0)}, \theta_2^{(0)})$ and $r = 1$.
Step 1. $\theta_2^{(r)} \leftarrow f(\theta_2 | \theta_1^{(r-1)})$.
Step 2. $\theta_1^{(r)} \leftarrow f(\theta_1 | \theta_2^{(r)})$.
Step 3. $r \leftarrow r + 1$.

Remarks ii

the transitional kernel is

$$K(\theta^{(r-1)}, \theta^{(r)}) = f(\theta_1^{(r)} | \theta_2^{(r)}) f(\theta_2^{(r)} | \theta_1^{(r-1)}).$$

$\boldsymbol{f(\theta)}$ is still the invariant density of this transition kernel.

2. The initial state $\boldsymbol{\theta}^{(0)}$ does not affect the convergence of the sample path as long as they are chosen from the support of $\boldsymbol{f(\theta)}$.

Example: Normal Distribution

Suppose $D = (x_1, \ldots, x_n)$ are independently sampled from $\mathbf{Normal}(\mu, \sigma^2)$. Assume the prior of μ is $\mathbf{Normal}(\mu_0, \tau_0^2)$ while that of σ^2 is $\mathbf{Inv.Gamma}\left(\frac{\nu_0}{2}, \frac{\lambda_0}{2}\right)$, which is not natural conjugate.

Gibbs sampler for $\mathbf{Normal}(\mu, \sigma^2)$

Step 1. $\mu^{(r)} \leftarrow$

$$\mathbf{Normal}\left(\frac{n(\sigma^{2(r-1)})^{-1}\bar{x} + \tau_0^{-2}\mu_0}{n(\sigma^{2(r-1)})^{-1} + \tau_0^{-2}}, \frac{1}{n(\sigma^{2(r-1)})^{-1} + \tau_0^{-2}}\right).$$

Step 2. $\sigma^{2(r)} \leftarrow$

$$\mathbf{Inv.Gamma}\left(\frac{n+\nu_0}{2}, \frac{n(\mu^{(r)} - \bar{x})^2 + \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda_0}{2}\right).$$

Proof Of $p(\mu|\sigma^2, D)$ i

The likelihood is

$$
\begin{aligned}
p(D|\mu, \sigma^2) &\propto (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{2\sigma^2}\right] \\
&\propto (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{n(\mu - \bar{x})^2 + \sum_{i=1}^{n}(x_i - \bar{x})^2}{2\sigma^2}\right],
\end{aligned}
$$

while the prior of $\boldsymbol{\mu}$ is

$$
p(\mu) = \frac{1}{\sqrt{2\pi\tau_0}} \exp\left[-\frac{(\mu - \mu_0)^2}{2\tau_0^2}\right].
$$

Proof Of $p(\mu|\sigma^2, D)$ ii

With Bayes' theorem, we have

$$
\begin{aligned}
&p(\mu|\sigma^2, D) \\
&\propto p(D|\mu, \sigma^2)p(\mu) \\
&\propto \mathbf{exp}\left[-\frac{n(\mu-\bar{x})^2 + \sum_{i=1}^{n}(x_i-\bar{x})^2}{2\sigma^2} - \frac{(\mu-\mu_0)^2}{2\tau_0^2}\right] \\
&\propto \mathbf{exp}\left[-\frac{n(\mu-\bar{x})^2}{2\sigma^2} - \frac{(\mu-\mu_0)^2}{2\tau_0^2}\right].
\end{aligned}
$$

Proof Of $p(\mu|\sigma^2, D)$ iii

By completing the square, we have

$$
\begin{aligned}
& n\sigma^{-2}(\mu - \bar{x})^2 + \tau_0^{-2}(\mu - \mu_0)^2 \\
&= (n\sigma^{-2} + \tau_0^{-2}) \left(\mu - \frac{n\sigma^{-2}\bar{x} + \tau_0^{-2}\mu_0}{n\sigma^{-2} + \tau_0^{-2}}\right)^2 + \text{constant}.
\end{aligned}
$$

Finally

$$
p(\mu|\sigma^2, D) \propto \mathbf{exp}\left[-\frac{n\sigma^{-2} + \tau_0^{-2}}{2}\left(\mu - \frac{n\sigma^{-2}\bar{x} + \tau_0^{-2}\mu_0}{n\sigma^{-2} + \tau_0^{-2}}\right)^2\right].
$$

∎

Proof Of $p(\sigma^2|\mu, D)$

With the prior of $\boldsymbol{\sigma}^2$

$$p(\sigma^2) = \frac{(\frac{\lambda_0}{2})^{\frac{\nu_0}{2}}}{\Gamma(\frac{\nu_0}{2})}(\sigma^2)^{-\left(\frac{\nu_0}{2}+1\right)} \exp\left[-\frac{\lambda_0}{2\sigma^2}\right],$$

we have

$$\begin{aligned} p(\sigma^2|\mu, D) &\propto p(D|\mu, \sigma^2)p(\sigma^2) \\ &\propto (\sigma^2)^{-\left(\frac{n+\nu_0}{2}+1\right)} \exp\left[-\frac{n(\mu-\bar{x})^2 + \sum_{i=1}^{n}(x_i-\bar{x})^2 + \lambda_0}{2\sigma^2}\right]. \end{aligned}$$

■

Gibbs Sampler For A Multivariate Distribution

Consider a k-variate distribution $f(\theta_1, \ldots, \theta_k)$ and suppose we can generate random numbers from the full conditional distribution of θ_j $(j = 1, \ldots, k)$:

$$f(\theta_j | \theta_1, \ldots, \theta_{j-1}, \theta_{j+1}, \ldots, \theta_k).$$

Gibbs sampler for the k-variate distribution

$$\begin{aligned}
\text{Step 1.}\quad & \theta_1^{(r)} \leftarrow f(\theta_1 | \theta_2^{(r-1)}, \ldots, \theta_k^{(r-1)}). \\
\text{Step 2.}\quad & \theta_2^{(r)} \leftarrow f(\theta_2 | \theta_1^{(r)}, \theta_3^{(r-1)}, \ldots, \theta_k^{(r-1)}). \\
& \vdots \\
\text{Step k-1.}\quad & \theta_{k-1}^{(r)} \leftarrow f(\theta_{k-1} | \theta_1^{(r)}, \ldots, \theta_{k-2}^{(r)}, \theta_k^{(r-1)}). \\
\text{Step k.}\quad & \theta_k^{(r)} \leftarrow f(\theta_k | \theta_1^{(r)}, \ldots, \theta_{k-1}^{(r)}).
\end{aligned}$$

Ex.: Truncated Multivariate Normal Distribution i

Consider a k-dimensional normal random vector $x = [x_1; \cdots; x_k] \sim \mathbf{Normal}(\mu, \Sigma)$ and impose $x_j > 0$, $(j = 1, \ldots, k)$. This follows a truncated multivariate normal distribution.

$$
\begin{aligned}
f(x|x > 0) &= \mathcal{K} \exp\left[-\frac{1}{2}(x - \mu)^\intercal \Sigma^{-1}(x - \mu)\right] \\
&\propto \exp\left[-\frac{1}{2}(x - \mu)^\intercal \Sigma^{-1}(x - \mu)\right] \prod_{j=1}^{k} \mathbb{1}_{(0,\infty)}(x_j),
\end{aligned}
$$

where $\mathcal{K} = (2\pi)^{-k/2}|\Sigma|^{-1/2}/\mathrm{P}(x_1 > 0, \ldots, x_k > 0)$.

Ex.: Truncated Multivariate Normal Distribution ii

To set up the Gibbs sampling algorithm, consider the following partition of $\boldsymbol{x}$, $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ as

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{x}_{-j} \\ x_j \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_{-j} \\ \mu_j \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{-j} & \boldsymbol{\sigma}_{-j}^{\intercal} \\ \boldsymbol{\sigma}_{-j} & \sigma_j^2 \end{bmatrix},$$

where "$-j$" indicates the part of $\boldsymbol{x}$ except for x_j.

Ex.: Truncated Multivariate Normal Distribution iii

Full conditional distribution of x_j

$$f(x_j | x_j > 0, x_{-j}) \propto \exp\left[-\frac{(x_j - \mu_j^*(x_{-j}))^2}{2\sigma_j^{2*}}\right] \mathbb{1}_{(0,\infty)}(x_j),$$

$$\mu_j^*(x_{-j}) = \mu_j + \sigma_{-j}^\intercal \Sigma_{-j}^{-1}(x_{-j} - \mu_{-j}),$$

$$\sigma_j^{2*} = \sigma_j^2 - \sigma_{-j}^\intercal \Sigma_{-j}^{-1} \sigma_{-j}.$$

This is a truncated normal distribution:

$$x_j | x_j > 0, x_{-j} \sim \text{Normal}^+(\mu_j^*(x_{-j}), \sigma_j^{2*}).$$

It is straightforward to generate random numbers from it.

Gibbs sampler for the truncated k-variate normal distribution

Step 1. $x_1^{(r)} \leftarrow \text{Normal}^+(\mu_1^*(x_2^{(r-1)}, \ldots, x_k^{(r-1)}), \sigma_1^{2*})$.

Step 2. $x_2^{(r)} \leftarrow \text{Normal}^+(\mu_2^*(x_1^{(r)}, x_3^{(r-1)}, \ldots, x_k^{(r-1)}), \sigma_2^{2*})$.

$\vdots$

Step k. $x_k^{(r)} \leftarrow \text{Normal}^+(\mu_k^*(x_1^{(r)}, \ldots, x_{k-1}^{(r)}), \sigma_k^{2*})$.

Consider the linear regression model:

$$y = X\beta + \epsilon, \quad \epsilon \sim \mathbf{Normal}(0, \sigma^2 \mathbf{I}).$$

Assume the prior of (β, σ^2) as

$$\beta \sim \mathbf{Normal}(\beta_0, A_0^{-1}), \quad \sigma^2 \sim \mathbf{Inv.Gamma}\left(\frac{\nu_0}{2}, \frac{\lambda_0}{2}\right),$$

which is not natural conjugate.

Linear Regression Model ii

The likelihood is

$$
\begin{aligned}
p(y|X,\theta) &= (2\pi\sigma^2)^{-\frac{n}{2}} \mathbf{exp}\left[-\frac{1}{2\sigma^2}(y - X\beta)^\intercal (y - X\beta)\right] \\
&\propto (\sigma^2)^{-\frac{n}{2}} \mathbf{exp}\Bigg[-\frac{1}{2\sigma^2}\{(y - X\hat{\beta})^\intercal (y - X\hat{\beta}) \\
&\qquad + (\beta - \hat{\beta})^\intercal X^\intercal X(\beta - \hat{\beta})\}\Bigg],
\end{aligned}
$$

where $\hat{\boldsymbol{\beta}}$ is the OLS estimator of $\boldsymbol{\beta}$, i.e.,

$$\hat{\beta} = (X^\intercal X)^{-1} X^\intercal y.$$

Linear Regression Model iii

With Bayes' theorem,

$$
\begin{aligned}
p(\beta, \sigma^2 | D) &\propto p(y|X, \theta) p(\beta) p(\sigma^2) \\
&\propto (\sigma^2)^{-\frac{n}{2}} \mathbf{exp}\Bigg[-\frac{1}{2\sigma^2}\Big\{(y - X\hat{\beta})^\intercal (y - X\hat{\beta}) \\
&\qquad + (\beta - \hat{\beta})^\intercal X^\intercal X(\beta - \hat{\beta})\Big\}\Bigg] \\
&\quad \times \mathbf{exp}\left[-\frac{1}{2}(\beta - \beta_0)^\intercal A_0 (\beta - \beta_0)\right] \\
&\quad \times (\sigma^2)^{-\left(\frac{\nu_0}{2}+1\right)} \mathbf{exp}\left[-\frac{\lambda_0}{2\sigma^2}\right].
\end{aligned}
$$

By completing the square, we have

$$
\begin{aligned}
&\sigma^{-2}(\beta - \hat{\beta})^\intercal X^\intercal X(\beta - \hat{\beta}) + (\beta - \beta_0)^\intercal A_0(\beta - \beta_0) \\
&= (\beta - \beta_\star)^\intercal \Sigma_\star^{-1}(\beta - \beta_\star) + \text{constant}, \\
&\beta_\star = (\sigma^{-2}X^\intercal X + A_0)^{-1}(\sigma^{-2}X^\intercal X\hat{\beta} + A_0\beta_0), \\
&\Sigma_\star = (\sigma^{-2}X^\intercal X + A_0)^{-1}.
\end{aligned}
$$

Thus we have the full conditional of $\boldsymbol{\beta}$ as

$$
p(\beta|\sigma^2, D) \propto \mathbf{exp}\left[-\frac{1}{2}(\beta - \beta_\star)^\intercal \Sigma_\star^{-1}(\beta - \beta_\star)^\intercal\right],
$$

which is $\mathbf{Normal}(\beta_\star, \Sigma_\star)$.

One the other hand, the full conditional of $\boldsymbol{\sigma}^2$ is derived as

$$p(\sigma^2|\beta, D) \propto (\sigma^2)^{-\left(\frac{\nu_\star}{2}+1\right)} \mathbf{exp}\left[-\frac{\lambda_\star}{2\sigma^2}\right],$$

$$\nu_\star = n + \nu_0,$$

$$\lambda_\star = (\beta - \hat{\beta})^\intercal X^\intercal X(\beta - \hat{\beta}) + (y - X\hat{\beta})^\intercal (y - X\hat{\beta}) + \lambda_0,$$

which is $\mathbf{Inv.Gamma}\left(\frac{\nu_\star}{2}, \frac{\lambda_\star}{2}\right)$.

Linear Regression Model vi

Gibbs sampler for the linear regression model

Step 1. $\beta^{(r)} \leftarrow \mathbf{Normal}\left(\beta_\star^{(r-1)}, \Sigma_\star^{(r-1)}\right)$.

Step 2. $\sigma^{2(r)} \leftarrow \mathbf{Inv.Gamma}\left(\frac{\nu_\star}{2}, \frac{\lambda_\star^{(r)}}{2}\right)$.

$\boldsymbol{\beta}_\star^{(r-1)}$ and $\boldsymbol{\Sigma}_\star^{(r-1)}$ depend on $\boldsymbol{\sigma}^{2(r-1)}$ generated in the previous cycle while $\boldsymbol{\lambda}_\star^{(r)}$ depends on $\boldsymbol{\beta}^{(r)}$ generated in Step 1.

Extensions Of Linear Regression Models

There are numerous ways to extend the normal linear regression model.

1. shrinkage prior
    - Ridge regression
    - LASSO
    - horseshoe prior
2. binary choice model
    - probit model
    - logit model
3. quantile regression
4. SUR model
5. panel data regression

Shrinkage Priors For Regression Coefficients

Consider the linear regression model:

$$y = X\beta + \epsilon, \quad \epsilon \sim \mathbf{Normal}(0, \sigma^2 \mathbf{I}).$$

Shrinkage priors are used to regularize the regression coefficients $\boldsymbol{\beta}$ when the number of independent variables/covariates is large. They are designed to shrink the regression coefficients toward zero, which is useful for variable selection and improving the prediction performance.

In this lecture, we will discuss three types of shrinkage prior: ridge regression, LASSO, and horseshoe prior.

Ridge regression is implemented by adding a penalty term $\lambda \sum_{j=1}^{k} \beta_j^2$ to the least squares problem as

$$\min_{\beta} \; (y - X\beta)^\intercal (y - X\beta) + \lambda \sum_{j=1}^{k} \beta_j^2,$$

which is equivalent to

$$\max_{\beta} \; -\frac{1}{2\sigma^2}(y - X\beta)^\intercal (y - X\beta) - \frac{1}{2\tau^2\sigma^2} \sum_{j=1}^{k} \beta_j^2, \; \tau = \sqrt{\frac{2}{\sigma^2 \lambda}}.$$

Ridge Regression ii

The first term in the objective function is interpreted as the log likelihood of the Gaussian linear regression model while the second is the log p.d.f. of a Gaussian distribution:

$$p(\beta_j) = \frac{1}{\sqrt{2\pi\tau^2\sigma^2}} \exp\left(-\frac{\beta_j^2}{2\tau^2\sigma^2}\right).$$

Therefore ridge regression is regarded as a MAP estimator of $\boldsymbol{\beta}$ with the Gaussian prior.

Prior For Ridge Regression

$$
\begin{aligned}
\beta_j|\tau^2, \sigma^2 &\sim \textbf{Normal}(0, \tau^2\sigma^2), \quad (j = 1, \ldots, k), \\
\tau^2 &\sim \textbf{Inv.Gamma}\left(\frac{a_\tau}{2}, \frac{b_\tau}{2}\right), \\
\sigma^2 &\sim \textbf{Inv.Gamma}\left(\frac{a_\sigma}{2}, \frac{b_\sigma}{2}\right).
\end{aligned}
$$

Marginalization With Respect To τ^2

Integrating out $\boldsymbol{\tau}^2$, we obtain the marginal prior of $\boldsymbol{\beta}_j$ as

$$p(\beta_j|\sigma) = \int_0^\infty p(\beta_j|\tau^2,\sigma^2)p(\tau^2)d\tau^2$$

$$\propto \int_0^\infty (\tau^2)^{-\frac{1}{2}} \mathbf{exp}\left(-\frac{\beta_j^2}{2\tau^2\sigma^2}\right)(\tau^2)^{-(\frac{a_\tau}{2}+1)} \mathbf{exp}\left(-\frac{b_\tau}{2\tau^2}\right) d\tau^2$$

$$\propto \int_0^\infty (\tau^2)^{-\left(\frac{a_\tau+1}{2}\right)} \mathbf{exp}\left(-\frac{\frac{\beta_j^2}{\sigma^2}+b_\tau}{2\tau^2}\right) d\tau^2 \propto \left(1+\frac{\beta_j^2}{\sigma^2 b_\tau}\right)^{-\frac{a_\tau+1}{2}},$$

which is a Student's $\boldsymbol{t}$ distribution with $\boldsymbol{a_\tau}$ degrees of freedom, location zero, and scale $\sqrt{\sigma^2 b_\tau / a_\tau}$.

LASSO (Least Absolute Shrinkage and Selection Operator) for the regression model is implemented by adding a penalty term $\lambda \sum_{j=1}^{k} |\beta_j|$ to the least squares problem as

$$\min_{\beta} \ (y - X\beta)^{\intercal} (y - X\beta) + \lambda \sum_{j=1}^{k} |\beta_j|,$$

which is equivalent to

$$\max_{\beta} \ -\frac{1}{2\sigma^2}(y - X\beta)^{\intercal} (y - X\beta) - \frac{1}{\tau\sigma} \sum_{j=1}^{k} |\beta_j|, \ \tau = \frac{2}{\sigma\lambda}.$$

LASSO ii

The first term in the objective function is interpreted as the log likelihood of the Gaussian linear regression model while the second is the log p.d.f. of a Laplace distribution:

$$p(\beta_j) = \frac{1}{2\tau\sigma} \exp\left(-\frac{|\beta_j|}{\tau\sigma}\right).$$

Therefore LASSO is regarded as a MAP estimator of $\boldsymbol{\beta}$ with the Laplace prior.

Scale-Mixture Expression Of A Laplace Distribution

A Laplace random variable $X \sim \mathbf{Laplace}(0, \sigma)$ is obtained with a scale-mixture of normal distributions:

$$X|W \sim \mathbf{Normal}(0, W\sigma^2), \quad W \sim \mathbf{Exponential}(2).$$

Prior For LASSO

$$
\begin{aligned}
\beta_j | \lambda_j^2, \tau^2, \sigma^2 &\sim \mathbf{Normal}(0, \lambda_j^2 \tau^2 \sigma^2), \quad (j = 1, \ldots, k), \\
\lambda_j^2 &\sim \mathbf{Exponential}(2), \\
\tau^2 &\sim \mathbf{Inv.Gamma}\left(\frac{a_\tau}{2}, \frac{b_\tau}{2}\right), \\
\sigma^2 &\sim \mathbf{Inv.Gamma}\left(\frac{a_\sigma}{2}, \frac{b_\sigma}{2}\right).
\end{aligned}
$$

Horseshoe Prior For Regression Coefficients

$$
\begin{aligned}
\beta_j|\lambda_j^2, \tau^2, \sigma^2 &\sim \mathbf{Normal}(0, \lambda_j^2\tau^2\sigma^2), \quad (j = 1, \ldots, k), \\
\lambda_j &\sim \mathbf{Cauchy}^+(0, 1), \\
\tau &\sim \mathbf{Cauchy}^+(0, 1),
\end{aligned}
$$

where $\mathbf{Cauchy}^+(0, 1)$ is the half-Cauchy distribution whose p.d.f. is

$$
f(x) = \frac{2}{\pi(1 + x^2)}, \quad x > 0.
$$

Mixture Expression Of A Half-Cauchy Distribution

A half-Cauchy random variable $X \sim \mathbf{Cauchy}^+(0, s)$ is obtained with a mixture of inverse gamma distributions:

$$X^2|W \sim \mathbf{Inv.Gamma}\left(\frac{1}{2}, \frac{1}{W}\right), \quad W \sim \mathbf{Inv.Gamma}\left(\frac{1}{2}, \frac{1}{s^2}\right).$$

Horseshoe Prior (Mixture Expression)

$$
\begin{aligned}
\beta_j|\lambda_j^2,\tau^2,\sigma^2 &\sim \mathbf{Normal}(0,\lambda_j^2\tau^2\sigma^2), \quad (j=1,\dots,k),\\
\lambda_j^2|\nu_j &\sim \mathbf{Inv.Gamma}\left(\frac{1}{2},\frac{1}{\nu_j}\right),\\
\tau^2|\xi &\sim \mathbf{Inv.Gamma}\left(\frac{1}{2},\frac{1}{\xi}\right),\\
\nu_1,\dots,\nu_k,\xi &\sim \mathbf{Inv.Gamma}\left(\frac{1}{2},1\right),\\
\sigma^2 &\sim \mathbf{Inv.Gamma}\left(\frac{a_\sigma}{2},\frac{b_\sigma}{2}\right).
\end{aligned}
$$

Conditional Prior of β

Recall that parameters in the condition are treated as fixed variables during Gibbs sampling. Once $\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_k$ and $\boldsymbol{\tau}$ are updated, the "conditional prior" of $\boldsymbol{\beta}$ is

$$\boldsymbol{\beta}|\boldsymbol{\theta}_{-\boldsymbol{\beta}} \sim \mathbf{Normal}(0, \boldsymbol{\sigma}^2 A^{-1}), \quad A^{-1} = \begin{bmatrix} \lambda_1^2\tau^2 & & 0 \\ & \ddots & \\ 0 & & \lambda_k^2\tau^2 \end{bmatrix},$$

where $\boldsymbol{\theta}$ is the set of all parameters and $\boldsymbol{\theta}_{-\boldsymbol{\beta}}$ is the set of parameters except for $\boldsymbol{\beta}$.

Note: This is virtually identical to the natural conjugate prior of $\boldsymbol{\beta}$ in the Gaussian linear regression model with the inverse gamma prior $\boldsymbol{\sigma}^2 \sim \mathbf{Inv.Gamma}\left(\frac{a_\sigma}{2}, \frac{b_\sigma}{2}\right)$.

Posterior Distribution

Applying Bayes' theorem, we have the posterior distribution:

$$
\begin{aligned}
p(\theta|D) &\propto p(D|\theta)p(\beta|\theta_{-\beta})p(\sigma^2)p(\theta_{-(\beta,\sigma^2)}) \\
&\propto (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2}(y - X\beta)^\intercal (y - X\beta)\right] \\
&\quad \times (\sigma^2)^{-\frac{k}{2}} |A|^{\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}\beta^\intercal A\beta\right) \\
&\quad \times (\sigma^2)^{-(\frac{a_\sigma}{2}+1)} \exp\left(-\frac{b_\sigma}{2\sigma^2}\right) \\
&\quad \times p(\theta_{-(\beta,\sigma^2)}),
\end{aligned}
$$

where $\theta_{-(\beta,\sigma^2)}$ is the set of parameters except for (β, σ^2).

The full conditional of $\boldsymbol{\beta}$ is

$$\boldsymbol{\beta}|\boldsymbol{\theta}_{-\boldsymbol{\beta}}, D \sim \textbf{Normal}\left(\boldsymbol{\beta}_\star, \sigma^2 A_\star^{-1}\right),$$
$$A_\star = X^\intercal X + A, \quad \boldsymbol{\beta}_\star = A_\star^{-1} X^\intercal y.$$

The full conditional of $\boldsymbol{\sigma}^2$ is

$$\sigma^2|\boldsymbol{\theta}_{-\sigma^2}, D \sim \textbf{Inv.Gamma}\left(\frac{a_\star}{2}, \frac{\bar{b}_\star}{2}\right),$$
$$a_\star = n + k + a_\sigma,$$
$$\bar{b}_\star = (y - X\beta)^\intercal (y - X\beta) + \beta^\intercal A \beta + b_\sigma.$$

Full Conditional Of (β, σ^2) ii

Since the conditional prior of $\boldsymbol{\beta}$ is virtually identical to the natural conjugate prior, we can jointly generate $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}^2$ by first drawing $\boldsymbol{\sigma}^2$ from

$$\boldsymbol{\sigma}^2 | \boldsymbol{\theta}_{-(\beta,\sigma^2)}, D \sim \textbf{Inv.Gamma}\left(\frac{a_\star}{2}, \frac{b_\star}{2}\right),$$

$$b_\star = (y - X\hat{\beta})^\intercal (y - X\hat{\beta}) + \hat{\beta}^\intercal \left((X^\intercal X)^{-1} + A^{-1}\right)^{-1} \hat{\beta} + b_\sigma,$$

and then drawing $\boldsymbol{\beta}$ from its full conditional.

Note that, unlike $(y - X\beta)^\intercal (y - X\beta)$, we need to compute $(y - X\hat{\beta})^\intercal (y - X\hat{\beta})$ only once before the Gibbs sampling.

Full Conditional Of τ^2 (Ridge Regression)

$$
\begin{aligned}
p(\tau^2 | \theta_{-\tau^2}, D) &\propto (\tau^2)^{-\frac{k}{2}} \exp\left(-\frac{\sum_{j=1}^{k} \beta_j^2}{2\tau^2\sigma^2}\right) \\
&\quad \times (\tau^2)^{-(\frac{a_\tau}{2}+1)} \exp\left(-\frac{b_\tau}{2\tau^2}\right) \\
&\propto (\tau^2)^{-\left(\frac{k+a_\tau}{2}+1\right)} \exp\left(-\frac{\frac{\sum_{j=1}^{k} \beta_j^2}{\sigma^2} + b_\tau}{2\tau^2}\right),
\end{aligned}
$$

which is $\textbf{Inv.Gamma}\left(\frac{k+a_\tau}{2}, \frac{1}{2}\left(\frac{\sum_{j=1}^{k}\beta_j^2}{\sigma^2} + b_\tau\right)\right)$.

Full Conditional Of λ_j^2 (LASSO) i

We have

$$
\begin{aligned}
p(\lambda_j^2 | \theta_{-\lambda_j^2}, D) &\propto (\lambda_j^2)^{-\frac{1}{2}} \exp\left(-\frac{\beta_j^2}{2\lambda_j^2 \tau^2 \sigma^2}\right) \times \exp\left(-\frac{\lambda_j^2}{2}\right) \\
&\propto (\lambda_j^2)^{-\frac{1}{2}} \exp\left[-\frac{1}{2}\left(\frac{\frac{\beta_j^2}{\tau^2\sigma^2}}{\lambda_j^2} + \lambda_j^2\right)\right].
\end{aligned}
$$

Full Conditional Of λ_j^2 (LASSO) ii

With the transformation $v_j = \lambda_j^{-2}$, we have

$$\begin{aligned} p(v_j|\theta_{-\lambda_j^2}, D) &\propto (v_j)^{\frac{1}{2}} \exp\left[-\frac{1}{2}\left(\frac{\beta_j^2}{\tau^2\sigma^2}v_j + \frac{1}{v_j}\right)\right] \times \frac{1}{v_j^2} \\ &\propto (v_j)^{-\frac{3}{2}} \exp\left[-\frac{1}{2}\left(\frac{\beta_j^2}{\tau^2\sigma^2}v_j + \frac{1}{v_j}\right)\right]. \end{aligned}$$

which is the inverse Gaussian distribution. Thus

$$\left.\frac{1}{\lambda_j^2}\right|\theta_{-\lambda_j^2}, D \sim \textbf{Inv.Gauss}\left(\frac{\tau\sigma}{|\beta_j|}, 1\right).$$

Inverse Gaussian Distribution

The p.d.f. of an inverse Gaussian distribution $\mathbf{Inv.Gauss}(\mu, \lambda)$ is

$$f(x) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp\left[-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right]$$

$$= \sqrt{\frac{b}{2\pi x^3}} \exp\left[-\sqrt{ab} - \frac{1}{2}\left(ax + \frac{b}{x}\right)\right],$$

$$\mu = \sqrt{\frac{b}{a}}, \quad \lambda = b.$$

Full Conditional Of τ^2 (LASSO)

$$
\begin{aligned}
p(\tau^2|\theta_{-\tau^2}, D) &\propto (\tau^2)^{-\frac{k}{2}} \exp\left(-\frac{1}{2\tau^2}\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}\right) \\
&\quad \times (\tau^2)^{-\left(\frac{a_\tau}{2}+1\right)} \exp\left(-\frac{b_\tau}{2\tau^2}\right) \\
&\propto (\tau^2)^{-\left(\frac{k+a_\tau}{2}+1\right)} \exp\left[-\frac{\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}+b_\tau}{2\tau^2}\right],
\end{aligned}
$$

which is $\textbf{Inv.Gamma}\left(\frac{k+a_\tau}{2}, \frac{1}{2}\left(\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}+b_\tau\right)\right)$.

Full Conditional Of λ_j^2 (Horseshoe)

$$
\begin{aligned}
p(\lambda_j^2 | \theta_{-\lambda_j^2}, D) &\propto (\lambda_j^2)^{-\frac{1}{2}} \exp\left(-\frac{\beta_j^2}{2\lambda_j^2 \tau^2 \sigma^2}\right) \\
&\quad \times (\lambda_j^2)^{-\left(\frac{1}{2}+1\right)} \exp\left(-\frac{1}{\nu_j \lambda_j^2}\right) \\
&\propto (\lambda_j^2)^{-(1+1)} \exp\left[-\frac{\frac{\beta_j^2}{2\tau^2\sigma^2} + \frac{1}{\nu_j}}{\lambda_j^2}\right],
\end{aligned}
$$

which is $\mathbf{Inv.Gamma}\left(1, \frac{\beta_j^2}{2\tau^2\sigma^2} + \frac{1}{\nu_j}\right)$.

Full Conditional Of τ^2 (Horseshoe)

$$
\begin{aligned}
p(\tau^2|\theta_{-\tau^2}, D) &\propto (\tau^2)^{-\frac{k}{2}} \exp\left(-\frac{1}{2\tau^2}\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}\right) \\
&\quad \times (\tau^2)^{-\left(\frac{1}{2}+1\right)} \exp\left(-\frac{1}{\xi\tau^2}\right) \\
&\propto (\tau^2)^{-\left(\frac{k+1}{2}+1\right)} \exp\left[-\frac{\frac{1}{2}\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}+\frac{1}{\xi}}{\tau^2}\right],
\end{aligned}
$$

which is $\textbf{Inv.Gamma}\left(\frac{k+1}{2}, \frac{1}{2}\sum_{j=1}^{k}\frac{\beta_j^2}{\lambda_j^2\sigma^2}+\frac{1}{\xi}\right)$.

Full Conditional Of ν_j (Horseshoe)

$$
\begin{aligned}
p(\nu_j|\theta_{-\nu_j}, D) &\propto \nu_j^{-\frac{1}{2}} \exp\left(-\frac{1}{\nu_j\lambda_j^2}\right) \times \nu_j^{-(\frac{1}{2}+1)} \exp\left(-\frac{1}{\nu_j}\right) \\
&\propto \nu_j^{-(1+1)} \exp\left(-\frac{\frac{1}{\lambda_j^2}+1}{\nu_j}\right),
\end{aligned}
$$

which is $\textbf{Inv.Gamma}\left(1, \frac{1}{\lambda_j^2}+1\right)$.

Full Conditional Of ξ (Horseshoe)

$$
\begin{aligned}
p(\xi|\theta_{-\xi}, D) &\propto \xi^{-\frac{1}{2}} \exp\left(-\frac{1}{\xi\tau^2}\right) \times \xi^{-\left(\frac{1}{2}+1\right)} \exp\left(-\frac{1}{\xi}\right) \\
&\propto \xi^{-(1+1)} \exp\left[-\frac{\frac{1}{\tau^2}+1}{\xi}\right],
\end{aligned}
$$

which is $\textbf{Inv.Gamma}\left(1, \frac{1}{\tau^2}+1\right)$.

Binary Dependent Variable

Suppose y_i takes either 1 or 0 with the probability:

$$y_i = \begin{cases} 1, & \text{with probability } q_i; \\ 0, & \text{with probability } 1 - q_i. \end{cases}$$

Unlike the Bernoulli distribution we covered before, the probability q_i differs from one individual to anther. This type of data appears in empirical analysis on decision making (e.g., consumer's choice) or events (e.g., bankruptcy).

Probit Model

Let x_i be a vector of characteristics of individual i ($i = 1, \ldots, n$) and suppose

$$q_i = F(x_i^\intercal \beta),$$

where F is the c.d.f. of a distribution. In practice, F is often supposed to be the c.d.f. of the standard normal distribution, i.e.,

$$F(x_i^\intercal \beta) = \int_{-\infty}^{x_i^\intercal \beta} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du.$$

Such a model is called the probit model.

Likelihood And Prior Distribution

The likelihood of the probit model is

$$
\begin{aligned}
p(y|\beta, X) &= \prod_{i=1}^{n} q_i^{y_i}(1 - q_i)^{1-y_i} \\
&= \prod_{i=1}^{n} F(x_i^\intercal \beta)^{y_i}(1 - F(x_i^\intercal \beta))^{1-y_i}.
\end{aligned}
$$

We suppose the prior for $\boldsymbol{\beta}$ is $\mathbf{Normal}(\boldsymbol{\beta_0}, \boldsymbol{A_0^{-1}})$.

Although it is not possible to derive the closed-form expression of the posterior distribution, we can construct a Gibbs sampling algorithm by introducing a latent variable related to the binary dependent variable y_i.

Probit Model As A Latent Variable Model

The probit model is equivalent to

$$z_i = x_i^\intercal \beta + \epsilon_i, \quad \epsilon_i \sim \mathbf{Normal}(0, 1),$$

$$y_i = \begin{cases} 1, & \text{if } z_i > 0; \\ 0, & \text{if } z_i \leqq 0. \end{cases}$$

Since

$$z_i \gtreqqless 0 \quad \Leftrightarrow \quad \epsilon_i \gtreqqless -x_i^\intercal \beta,$$

we have

$$\mathbf{P}(y_i = 1) = \mathbf{P}(\epsilon_i > -x_i^\intercal \beta) = \mathbf{P}(-\epsilon_i < x_i^\intercal \beta) = F(x_i^\intercal \beta),$$

$$\mathbf{P}(y_i = 0) = \mathbf{P}(\epsilon_i \leqq -x_i^\intercal \beta) = \mathbf{P}(-\epsilon_i \geqq x_i^\intercal \beta) = 1 - F(x_i^\intercal \beta).$$

Likelihood With Latent Variables i

Note that

$$\mathbf{P}(y_i = 1) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(z_i - x_i^\intercal \beta)^2}{2}\right] \mathbb{1}^+(z_i) dz_i,$$

$$\mathbf{P}(y_i = 0) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(z_i - x_i^\intercal \beta)^2}{2}\right] \mathbb{1}^-(z_i) dz_i,$$

where $\mathbb{1}^+(z_i)$ takes one if z_i is positive; otherwise it takes zero, and $\mathbb{1}^-(z_i) = 1 - \mathbb{1}^+(z_i)$.

Likelihood With Latent Variables ii

Let us pretend $\boldsymbol{z} = [z_1; \cdots; z_n]$ are observed. Then the likelihood is rewritten as

$$\begin{aligned} & p(y, z|\beta, X) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(z_i - x_i^\intercal \beta)^2}{2}\right] \left\{y_i \mathbb{1}^+(z_i) + (1 - y_i)\mathbb{1}^-(z_i)\right\}. \end{aligned}$$

Note that

$$p(y|\beta, X) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(y, z|\beta, X) dz_1 \cdots dz_n.$$

We will derive the full conditional posterior distributions with the above likelihood.

Data Augmentation i

The posterior distribution of $\boldsymbol{\beta}$ is

$$p(\beta|D) = \frac{p(y|\beta, X)p(\beta)}{\int p(y|\beta, X)p(\beta)d\beta}$$
$$= \frac{\int p(y, z|\beta, X)p(\beta)dz}{\int \int p(y, z|\beta, X)p(\beta)dzd\beta}$$
$$= \int p(\beta, z|D)dz,$$

where

$$p(\beta, z|D) = \frac{p(y, z|\beta, X)p(\beta)}{\int \int p(y, z|\beta, X)p(\beta)dzd\beta}.$$

Data Augmentation ii

Thus, by generating $\boldsymbol{\beta}$ and $\boldsymbol{z}$ jointly from $\boldsymbol{p(\beta, z|D)}$ and keeping only $\boldsymbol{\beta}$, we can obtain the Monte Carlo sample of $\boldsymbol{\beta}$ from the posterior distribution $\boldsymbol{p(\beta|D)}$. This approach is called data augmentation. The algorithm is similar to Gibbs sampling.

Data augmentation

Step 0. Set the initial $\boldsymbol{\beta}^{(0)}$ and $\boldsymbol{z}^{(0)}$ and $r \leftarrow 1$.

Step 1. $\beta^{(r+1)} \leftarrow p(\beta|z^{(r)}, D)$.

Step 2. $z^{(r+1)} \leftarrow p(z|\beta^{(r+1)}, D)$.

Step 3. $r \leftarrow r + 1$ and go to Step 1.

Full Conditional Of β

Since the likelihood with latent variables is none other than that of the linear regression model given $\boldsymbol{z}$, the full conditional of $\boldsymbol{\beta}$ is derived as

$$\beta | z, D \sim \mathbf{Normal}(\beta_\star, \Sigma_\star),$$

where

$$\beta_\star = (X^\intercal X + A_0)^{-1}(X^\intercal z + A_0\beta_0), \quad \Sigma_\star = (X^\intercal X + A_0)^{-1}.$$

Full Conditional Of The Latent Variable

The full conditional of z_i $(i = 1, \ldots, n)$ is the following truncated normal distribution:

$$z_i | \beta, D \sim \begin{cases} \mathbf{Normal}^+(x_i^\intercal \beta, 1), & \text{if } y_i = 1; \\ \mathbf{Normal}^-(x_i^\intercal \beta, 1), & \text{if } y_i = 0, \end{cases}$$

where $\mathbf{Normal}^+$ and $\mathbf{Normal}^-$ stand for positively and negatively truncated normal distribution respectively.

Logit Model i

Suppose the latent variable follows the standard logistic distribution with the c.d.f.: $F(z) = e^z/(1+e^z)$. This means

$$\mathbf{P}(y_i = 1) = \frac{\exp(x_i^\intercal \beta)}{1+\exp(x_i^\intercal \beta)}, \quad \mathbf{P}(y_i = 0) = \frac{1}{1+\exp(x_i^\intercal \beta)},$$

which is call the logit model. To derive a Gibbs sampler for this model, we consider a Pólya-gamma distribution:

$$X = \frac{1}{2\pi^2}\sum_{k=1}^{\infty} \frac{g_k}{\left(k-\frac{1}{2}\right)^2 + \frac{c^2}{(4\pi)^2}} \sim \mathbf{PG}(b, c),$$

where g_k independently follows $\mathbf{Gamma}(b, 1)$.

Logit Model ii

According to Polson et al. (2013),

$$\frac{\left(e^{\psi}\right)^a}{\left(1+e^{\psi}\right)^b} = \frac{1}{2^b} e^{\kappa\psi} \int_0^\infty e^{-\frac{\omega\psi^2}{2}} p(\omega) d\omega,$$

where $\kappa = a - b/2$ and $\omega \sim \mathbf{PG}(b, 0)$. Since

$$\frac{\left(e^{\psi}\right)^a}{\left(1+e^{\psi}\right)^b} = \begin{cases} \dfrac{e^{\psi}}{1+e^{\psi}} & (a=1, b=1); \\ \dfrac{1}{1+e^{\psi}} & (a=0, b=1), \end{cases}$$

we may apply a data augmentation scheme for the logit model by generating ω from its full conditional.

Logit Model iii

Polson et al. (2013) shows that the full conditional of $\boldsymbol{\omega}$ given $\boldsymbol{\psi}$ is $\boldsymbol{\omega}|\boldsymbol{\psi} \sim \mathbf{PG}(b, \boldsymbol{\psi})$. Since the likelihood is

$$p(D|\beta) = \prod_{i=1}^{n} \frac{\left(\exp(x_i^\intercal \beta)\right)^{y_i}}{1 + \exp(x_i^\intercal \beta)},$$

it is rewritten as

$$p(D|\beta) = \prod_{i=1}^{n} \frac{1}{2} \exp(\kappa_i x_i^\intercal \beta) \int_0^\infty \exp\left(-\frac{\omega \beta^\intercal x_i x_i^\intercal \beta}{2}\right) p(\omega) d\omega,$$

where $\kappa_i = y_i - \frac{1}{2}$.

Logit Model iv

Furthermore, defining $\boldsymbol{\omega} = [\omega_1; \cdots; \omega_n]$, we have

$$
\begin{aligned}
p(D|\beta) &= \int_0^\infty \cdots \int_0^\infty p(D|\beta, \omega) p(\omega_1) d\omega_1 \cdots p(\omega_n) d\omega_n \\
&= \int_0^\infty \cdots \int_0^\infty \frac{1}{2^n} \exp\left[\left(\sum_{i=1}^n \kappa_i x_i^\top\right)\beta - \frac{1}{2}\beta^\top \left(\sum_{i=1}^n \omega_i x_i x_i^\top\right)\beta\right] \\
&\qquad \times p(\omega_1) d\omega_1 \cdots p(\omega_n) d\omega_n \\
&= \int_0^\infty \cdots \int_0^\infty \frac{1}{2^n} \exp\left(\kappa^\top X\beta - \frac{1}{2}\beta^\top X^\top \boldsymbol{\Omega} X\beta\right) \\
&\qquad \times p(\omega_1) d\omega_1 \cdots p(\omega_n) d\omega_n,
\end{aligned}
$$

where $\boldsymbol{\kappa} = [\kappa_1; \cdots; \kappa_n]$ and $\boldsymbol{\Omega} = \mathbf{diag}(\boldsymbol{\omega})$.

Treating $\boldsymbol{\omega}$ as latent variables, the posterior distribution of $(\boldsymbol{\beta}, \boldsymbol{\omega})$ is derived as

$$\begin{aligned} p(\boldsymbol{\beta}, \boldsymbol{\omega} | D) &\propto p(D | \boldsymbol{\beta}, \boldsymbol{\omega}) p(\boldsymbol{\beta}) p(\boldsymbol{\omega}) \\ &= \exp\left(\boldsymbol{\kappa}^\intercal X \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\intercal X^\intercal \boldsymbol{\Omega} X \boldsymbol{\beta}\right) \\ &\quad \times \exp\left[-\frac{1}{2} (\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\intercal A_0 (\boldsymbol{\beta} - \boldsymbol{\beta}_0)\right] p(\boldsymbol{\omega}), \end{aligned}$$

where we suppose $p(\boldsymbol{\beta})$ is $\mathbf{Normal}(\boldsymbol{\beta}_0, A_0^{-1})$.

The full conditionals are

$$
\begin{aligned}
\beta | \omega, D &\sim \mathbf{Normal}(\beta_\star, \Sigma_\star), \\
\omega_i | \beta, D &\sim \mathbf{PG}(1, x_i^\intercal \beta), \quad (i = 1, \ldots, n),
\end{aligned}
$$

where

$$
\beta_\star = (X^\intercal \Omega X + A_0)^{-1}(X^\intercal \kappa + A_0 \beta_0), \quad \Sigma_\star = (X^\intercal \Omega X + A_0)^{-1}.
$$

Quantile Regression

The α-quantile of a random variable y, q_α, is defined as

$$\mathbf{P}(y \leqq q_\alpha) = \alpha, \quad 0 \leqq \alpha \leqq 1.$$

Suppose q_α is expressed as a linear combination of x, i.e.,

$$\mathbf{P}(y \leqq x^\intercal \beta_\alpha) = \alpha.$$

Then a quantile regression (QR) is formulated as

$$y = x^\intercal \beta_\alpha + \epsilon, \quad \int_{-\infty}^{0} f(\epsilon) d\epsilon = \alpha,$$

where f is the p.d.f. of the error term ϵ.

Quantile Regression Estimation

The coefficients $\boldsymbol{\beta}$ in QR are estimated by solving the following minimization problem:

$$\min_{\beta_\alpha} \sum_{i=1}^{n} \rho_\alpha \left(y_i - x_i^\intercal \beta_\alpha \right),$$

where $\boldsymbol{\rho_\alpha(\cdot)}$ is a type of loss function (check function):

$$\rho_\alpha(u) = u\{\alpha - \mathbb{1}^-(u)\}.$$

Asymmetric Laplace Likelihood

Since

$$\rho_\alpha(y_i - x_i^\intercal \beta_\alpha) = (y_i - x_i^\intercal \beta_\alpha)\{\alpha - \mathbb{1}^-(y_i - x_i^\intercal \beta_\alpha)\}$$
$$= \begin{cases} \alpha(y_i - x_i^\intercal \beta_\alpha), & \text{if } y_i > x_i^\intercal \beta_\alpha; \\ (\alpha - 1)(y_i - x_i^\intercal \beta_\alpha), & \text{if } y_i \leqq x_i^\intercal \beta_\alpha, \end{cases}$$

QR is equivalent to

$$\max_{\beta_\alpha} \prod_{i=1}^{n} \frac{\alpha(1-\alpha)}{\sigma} \exp\Big[-\frac{\alpha}{\sigma}(y_i - x_i^\intercal \beta_\alpha)\mathbb{1}^+(y_i - x_i^\intercal \beta_\alpha) - \frac{\alpha - 1}{\sigma}(y_i - x_i^\intercal \beta_\alpha)\mathbb{1}^-(y_i - x_i^\intercal \beta_\alpha)\Big],$$

which is the maximum likelihood estimation of the asymmetric Laplace distribution.

Mixture Representation Of Asymmetric Laplace i

Let us rearrange the asymmetric Laplace likelihood to derive a Gibbs sampling algorithm for QR.

The asymmetric Laplace distribution can be expressed as a mixture of normal distributions:

$$
\begin{aligned}
& y_i = x_i^\intercal \beta_\alpha + \theta z_i + \tau \sqrt{\sigma z_i} u_i, \\
& z_i \sim \textbf{Exponential}(\sigma), \quad u_i \sim \textbf{Normal}(0, 1), \; z_i \perp u_i, \\
& \theta = \frac{1 - 2\alpha}{\alpha(1-\alpha)}, \quad \tau^2 = \frac{2}{\alpha(1-\alpha)}.
\end{aligned}
$$

Note that θ and τ are fixed once α is given.

Mixture Representation Of Asymmetric Laplace ii

Then the likelihood for data augmentation is

$$\begin{aligned} & p(y, z|\boldsymbol{\beta}_\alpha, \boldsymbol{\sigma}, X) \\ & \propto \prod_{i=1}^{n} \frac{1}{\sqrt{\boldsymbol{\sigma} z_i}} \exp\left[-\frac{(y_i - x_i^\intercal \boldsymbol{\beta}_\alpha - \theta z_i)^2}{2\boldsymbol{\tau}^2 \boldsymbol{\sigma} z_i}\right] \\ & \quad \times \prod_{i=1}^{n} \frac{1}{\boldsymbol{\sigma}} \exp\left(-\frac{z_i}{\boldsymbol{\sigma}}\right). \end{aligned}$$

We suppose the following priors for $\boldsymbol{\beta}_\alpha$ and $\boldsymbol{\sigma}$.

$$\boldsymbol{\beta}_\alpha \sim \mathbf{Normal}(\boldsymbol{\beta}_0, A_0^{-1}), \quad \boldsymbol{\sigma} \sim \mathbf{Inv.Gamma}\left(\frac{\nu_0}{2}, \frac{\lambda_0}{2}\right).$$

The full conditional of $\boldsymbol{\beta}_\alpha$ is

$$\boldsymbol{\beta}_\alpha|\boldsymbol{\sigma}, z, D \sim \textbf{Normal}\left(\boldsymbol{\beta}_\star, A_\star^{-1}\right),$$

$$A_\star = \sum_{i=1}^{n} \frac{x_i x_i^\intercal}{\tau^2 \sigma z_i} + A_0,$$

$$\boldsymbol{\beta}_\star = A_\star^{-1}\left(\sum_{i=1}^{n} \frac{x_i(y_i - \theta z_i)}{\tau^2 \sigma z_i} + A_0 \boldsymbol{\beta}_0\right).$$

The full conditional of $\boldsymbol{\sigma}$ is

$$\boldsymbol{\sigma}|\boldsymbol{\beta}_\alpha, z, D \sim \textbf{Inv.Gamma}\left(\frac{\nu_\star}{2}, \frac{\lambda_\star}{2}\right),$$

$$\nu_\star = 3n + \nu_0, \quad \lambda_\star = \sum_{i=1}^{n} \frac{(y_i - x_i^\intercal \boldsymbol{\beta}_\alpha - \theta z_i)^2}{\tau^2 z_i} + 2\sum_{i=1}^{n} z_i + \lambda_0.$$

Full Conditionals ii

The full conditional of z_i is

$$p(z_i|\beta_\alpha, \sigma, D) \propto z_i^{-\frac{1}{2}} \exp\left[-\frac{(y_i - x_i^\intercal \beta_\alpha - \theta z_i)^2}{2\tau^2\sigma z_i} - \frac{z_i}{\sigma}\right]$$
$$\propto z_i^{-\frac{1}{2}} \exp\left[-\frac{1}{2}\left(\hat{a}_i z_i + \frac{\hat{b}_i}{z_i}\right)\right],$$

where

$$\hat{a}_i = \frac{\theta^2}{\tau^2\sigma} + \frac{2}{\sigma}, \quad \hat{b}_i = \frac{(y_i - x_i^\intercal \beta_\alpha)^2}{\tau^2\sigma}.$$

This is the generalized inverse Gaussian distribution $\textbf{Gen.Inv.Gauss}\left(\frac{1}{2}, \hat{a}_i, \hat{b}_i\right)$.

Full Conditionals iii

In general, the p.d.f. of $\mathbf{Gen.Inv.Gauss}(p, a, b)$ is

$$f(x|p,a,b) = \frac{\left(\frac{a}{b}\right)^{\frac{p}{2}}}{2K_p(\sqrt{ab})} x^{p-1} \exp\left[-\frac{1}{2}\left(ax + \frac{b}{x}\right)\right],$$

where $K_p(\cdot)$ is the modified Bessel function of the second kind. By defining $\xi = \sqrt{ab}$ and $\eta = \sqrt{b/a}$, the generalized inverse gaussian distribution is expressed as

$$f(x|p,\xi,\eta) = \frac{1}{2\eta K_p(\xi)} \left(\frac{x}{\eta}\right)^{p-1} \exp\left[-\frac{\xi}{2}\left(\frac{x}{\eta} + \frac{\eta}{x}\right)\right].$$

η is regarded as the scale parameter and the distribution is in the standard form when $\eta = 1$.

SUR Model

A seemingly unrelated regression (SUR) model is a system of N linear regression model:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} X_1 & & & \\ & X_2 & & \\ & & \ddots & \\ & & & X_N \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_N \end{bmatrix},$$

with correlated error terms:

$$\mathbf{E}[\epsilon_i \epsilon_j^\intercal] = \begin{cases} \sigma_i^2 \mathbf{I}, & (i = j); \\ \sigma_{ij} \mathbf{I}, & (i \neq j). \end{cases}$$

Let $y = [y_1; \cdots; y_N]$ and $Z = \mathrm{diag}\{X_1, \ldots, X_N\}$. Then

$$y = Z\beta + \epsilon, \quad \epsilon \sim \mathbf{Normal}(0, \Sigma \otimes \mathbf{I}).$$

Likelihood

The likelihood of the SUR model is

$$
\begin{aligned}
& p(y|\beta, \Sigma, Z) \\
& \propto |\Sigma|^{-\frac{T}{2}} \mathbf{exp} \left[-\frac{1}{2} (y - Z\beta)^{\intercal} (\Sigma^{-1} \otimes \mathbf{I}) (y - Z\beta) \right] \\
& \propto |\Sigma|^{-\frac{T}{2}} \mathbf{exp} \left[-\frac{1}{2} \mathrm{tr}(\Sigma^{-1} S) \right],
\end{aligned}
$$

where

$$
S = \begin{bmatrix} e_1^{\intercal} e_1 & \cdots & e_1^{\intercal} e_N \\ \vdots & \ddots & \vdots \\ e_N^{\intercal} e_1 & \cdots & e_N^{\intercal} e_N \end{bmatrix}, \; e_i = y_i - X_i \beta_i, \; (i = 1, \ldots, N).
$$

Prior Distributions

We suppose the priors for $\boldsymbol{\beta}$ and $\boldsymbol{\Sigma}$ are

$$\boldsymbol{\beta} \sim \mathbf{Normal}(\boldsymbol{\beta}_0, A_0^{-1}), \quad \boldsymbol{\Sigma} \sim \mathbf{Inv.Wishart}(\bar{\boldsymbol{\Lambda}}, \nu_0).$$

An inverse Wishart distribution $\mathbf{Inv.Wishart}(\boldsymbol{\Lambda}, \nu)$ is

$$p(\boldsymbol{\Sigma}|\boldsymbol{\Lambda}, \nu) = \frac{|\boldsymbol{\Lambda}|^{\frac{\nu}{2}}}{2^{\frac{\nu m}{2}} \boldsymbol{\Gamma}_m\left(\frac{\nu}{2}\right)} |\boldsymbol{\Sigma}|^{-\frac{\nu+m+1}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda})\right],$$

where

$$\boldsymbol{\Gamma}_m(u) = \pi^{\frac{m(m-1)}{4}} \prod_{j=1}^{m} \boldsymbol{\Gamma}\left(u + \frac{1-j}{2}\right).$$

By defining

$$
\begin{aligned}
\tilde{\beta} &= \left\{ Z^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) Z \right\}^{-1} Z^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) y, \\
\tilde{\boldsymbol{\Sigma}} &= \left\{ Z^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) Z \right\}^{-1}, \\
\tilde{s}^2 &= (y - Z\tilde{\beta})^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) (y - Z\tilde{\beta}) \\
&= y^{\intercal} \Big[(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) - (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) Z \left\{ Z^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) Z \right\}^{-1} \\
&\qquad\qquad \times Z^{\intercal} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}) \Big] y,
\end{aligned}
$$

Posterior Distribution ii

we have

$$(y - Z\beta)^\intercal(\Sigma^{-1} \otimes \mathrm{I})(y - Z\beta) = \tilde{s}^2 + (\beta - \tilde{\beta})^\intercal \tilde{\Sigma}^{-1}(\beta - \tilde{\beta}).$$

Completing the square, we have

$$\begin{aligned}(y - Z\beta)^\intercal(\Sigma^{-1} \otimes \mathrm{I})(y - Z\beta) &+ (\beta - \beta_0)^\intercal A_0(\beta - \beta_0) \\ &= (\beta - \beta_\star)^\intercal \Sigma_\star^{-1}(\beta - \beta_\star) + \tilde{s}^2 + (\beta_0 - \tilde{\beta})^\intercal \Omega_\star^{-1}(\beta_0 - \tilde{\beta}),\end{aligned}$$

where

$$\beta_\star = (\tilde{\Sigma}^{-1} + A_0)^{-1}(\tilde{\Sigma}^{-1}\tilde{\beta} + A_0\beta_0), \quad \Sigma_\star = (\tilde{\Sigma}^{-1} + A_0)^{-1},$$
$$\Omega_\star = \tilde{\Sigma} + A_0^{-1}.$$

Therefore we can rewrite the posterior distribution as

$$p(\beta, \Sigma|D) \propto |\Sigma|^{-\frac{T+\nu_0+N+1}{2}} \exp\left[-\frac{1}{2}(\beta-\beta_\star)^\intercal \Sigma_\star^{-1}(\beta-\beta_\star)\right]$$
$$\times \exp\left[-\frac{1}{2}\left\{\mathrm{tr}(\Sigma^{-1}\bar{\Lambda}) + \tilde{s}^2 + (\beta_0-\tilde{\beta})^\intercal \Omega_\star^{-1}(\beta_0-\tilde{\beta})\right\}\right].$$

Thus we have the full conditional posterior distribution of $\boldsymbol{\beta}$ as

$$p(\beta|\Sigma, D) \propto \exp\left[-\frac{1}{2}(\beta - \beta_\star)^\intercal \Sigma_\star^{-1}(\beta - \beta_\star)\right],$$

which is $\mathbf{Normal}(\boldsymbol{\beta}_\star, \boldsymbol{\Sigma}_\star)$.

Full Conditional Of Σ

Alternatively, using the second expression of the likelihood, we have

$$p(\beta, \Sigma | D) \propto |\Sigma|^{-\frac{T+\nu_0+N+1}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left\{\Sigma^{-1}(S+\bar{\Lambda})\right\}\right]$$
$$\times \exp\left[-\frac{1}{2}(\beta-\beta_0)^\intercal A_0(\beta-\beta_0)\right].$$

Thus we have the full conditional posterior distribution of Σ as

$$p(\Sigma | \beta, D) \propto |\Sigma|^{-\frac{\nu_\star+N+1}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\Sigma^{-1}\Lambda_\star\right)\right],$$

where $\Lambda_\star = S + \bar{\Lambda}$ and $\nu_\star = T + \nu_0$.

This is $\mathbf{Inv.Wishart}(\Lambda_\star, \nu_\star)$.

Panel Data

In a typical panel data set, each observation y_{it} has two indices: $i = 1, \dots, N$ and $t = 1, \dots, T$. The former corresponds to an entity (person, household, firm, $\dots$) while the latter indicates a time period (day, month, year, $\dots$). Panel data can be tabulated as

$$
\begin{array}{c|ccccc}
t \backslash i & 1 & \cdots & i & \cdots & N \\
\hline
1 & y_{11} & & \cdots & & y_{N1} \\
\vdots & & \ddots & & & \\
t & \vdots & & y_{it} & & \vdots \\
\vdots & & & & \ddots & \\
T & y_{1T} & & \cdots & & y_{NT}
\end{array}
$$

Pooled Regression And Unobserved Heterogeneity

A pooled regression of panel data is a typical linear regression such that

$$y_{it} = \alpha + x_{it}^{\intercal}\beta + \epsilon_{it}, \quad \epsilon_{it} \sim \mathrm{Normal}(0, \sigma^2),$$
$$(i = 1, \ldots, N, \; t = 1, \ldots, T),$$

where x_{it} does not include the constant term.

In reality, the conditional expectation of y_{it} is often affected by an unobserved factor, which is referred to as the unobserved heterogeneity.

Panel Data Regression

In order to deal with the unobserved heterogeneity, we introduce the individual-specific constant term α_i $(i = 1, \ldots, N)$, that is,

$$y_{it} = \alpha_i + x_{it}^\top \beta + \epsilon_{it}, \quad \epsilon_{it} \sim \mathbf{Normal}(0, \sigma^2),$$

which is called the panel data regression model.

Frequentist Formulations Of Panel Data Regression

(a) Fixed-Effects (FE) Model

α_i is supposed to be a fixed unknown parameter.

$$y_{it} = \sum_{j=1}^{N} \alpha_j \mathbb{1}_j(i) + x_{it}^\intercal \beta + \epsilon_{it}, \quad \epsilon_{it} \sim \mathbf{Normal}(0, \sigma^2).$$

(b) Random-Effects (RE) Model

α_i is a individual-specific error term.

$$y_{it} = \alpha_0 + x_{it}^\intercal \beta + u_i + \epsilon_{it},$$
$$\epsilon_{it} \sim \mathbf{Normal}(0, \sigma^2), \quad u_i \sim \mathbf{Normal}(0, \tau^2),$$

where $\alpha_i = \alpha_0 + u_i$.

Bayesian Interpretation Of Panel Data Regression

From the Bayesian perspective, the panel data regression model is a type of hierarchical Bayes model in which the prior distribution

$$\alpha_i \sim \mathrm{Normal}(\alpha_0, \tau^2), \quad (i = 1, \dots, N),$$

is used for $\alpha_1, \dots, \alpha_N$. In this context, no fundamental distinction exists between FE model and RE model.

- For the FE model, it is the common prior for all α_i's.
- For the RE model, it is the distribution of the individual-specific error term.

Prior Distributions

We suppose the following prior distributions:

$$\begin{aligned}
\boldsymbol{\beta} &\sim \mathbf{Normal}(\boldsymbol{\beta}_0, A_0^{-1}),\\
\boldsymbol{\sigma}^2 &\sim \mathbf{Inv.Gamma}\left(\frac{\boldsymbol{\nu}_{\boldsymbol{\sigma}}}{2}, \frac{\boldsymbol{\lambda}_{\boldsymbol{\sigma}}}{2}\right),\\
\boldsymbol{\alpha}_0 &\sim \mathbf{Normal}(\boldsymbol{\mu}, \boldsymbol{\xi}^2),\\
\boldsymbol{\tau}^2 &\sim \mathbf{Inv.Gamma}\left(\frac{\boldsymbol{\nu}_{\boldsymbol{\tau}}}{2}, \frac{\boldsymbol{\lambda}_{\boldsymbol{\tau}}}{2}\right).
\end{aligned}$$

Likelihood

The likelihood of the panel data regression model is

$$p(y|X,\theta) \propto (\sigma^2)^{-\frac{NT}{2}} \,\mathbf{exp}\left[-\frac{1}{2\sigma^2}(y - D\alpha - X\beta)^\intercal (y - D\alpha - X\beta)\right],$$

where

$$y = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1T} \\ \vdots \\ y_{N1} \\ \vdots \\ y_{NT} \end{bmatrix}, \quad X = \begin{bmatrix} x_{11}^\intercal \\ \vdots \\ x_{1T}^\intercal \\ \vdots \\ x_{N1}^\intercal \\ \vdots \\ x_{NT}^\intercal \end{bmatrix}, \quad D = \begin{bmatrix} \boldsymbol{\iota} & & \\ & \ddots & \\ & & \boldsymbol{\iota} \end{bmatrix}, \quad \alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{bmatrix},$$

and $\boldsymbol{\iota}$ is a $T \times 1$ vector of 1's.

Full Conditional Of β

With Bayes' theorem, we have

$$p(\boldsymbol{\beta}|\boldsymbol{\theta}_{-\boldsymbol{\beta}}, \text{data}) \propto \exp\left[-\frac{1}{2\sigma^2}(\tilde{y}_\star - X\boldsymbol{\beta})^\intercal(\tilde{y}_\star - X\boldsymbol{\beta})\right] \\ \times \exp\left[-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\intercal A_0(\boldsymbol{\beta} - \boldsymbol{\beta}_0)\right],$$

where $\tilde{y}_\star = y - D\boldsymbol{\alpha}$. Thus the full conditional of $\boldsymbol{\beta}$ is

$$\boldsymbol{\beta}|\boldsymbol{\theta}_{-\boldsymbol{\beta}}, \text{data} \sim \mathbf{Normal}(\boldsymbol{\beta}_\star, \boldsymbol{\Sigma}_\star),$$

where $\boldsymbol{\beta}_\star = \boldsymbol{\Sigma}_\star\left(\sigma^{-2}X^\intercal\tilde{y}_\star + A_0\boldsymbol{\beta}_0\right)$ and
$\boldsymbol{\Sigma}_\star = \left(\sigma^{-2}X^\intercal X + A_0\right)^{-1}$.

Full Conditional Of α

In the same fashion as $\boldsymbol{\beta}$, we have

$$\begin{aligned} p(\boldsymbol{\alpha}|\boldsymbol{\theta}_{-\boldsymbol{\alpha}}, \text{data}) \propto \mathbf{exp}\left[-\frac{1}{2\sigma^2}(\boldsymbol{e} - \boldsymbol{D\alpha})^\intercal(\boldsymbol{e} - \boldsymbol{D\alpha})\right] \\ \times \mathbf{exp}\left[-\frac{1}{2\tau^2}(\boldsymbol{\alpha} - \alpha_0\boldsymbol{\iota})^\intercal(\boldsymbol{\alpha} - \alpha_0\boldsymbol{\iota})\right], \end{aligned}$$

where $\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X\beta}$. Thus the full conditional of $\boldsymbol{\alpha}$ is

$$\boldsymbol{\alpha}|\boldsymbol{\theta}_{-\boldsymbol{\alpha}}, \text{data} \sim \mathbf{Normal}(\boldsymbol{\alpha}_\star, \boldsymbol{\Omega}_\star),$$

where $\boldsymbol{\alpha}_\star = \boldsymbol{\Omega}_\star\left(\sigma^{-2}\boldsymbol{D}^\intercal\boldsymbol{e} + \frac{\alpha_0}{\tau^2}\boldsymbol{\iota}\right)$ and $\boldsymbol{\Omega}_\star = \left(\frac{T}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}\mathbf{I}$.

Full Conditionals Of σ^2, α_0 and τ^2

$$\boldsymbol{\sigma}^2|\boldsymbol{\theta}_{-\sigma^2}, \text{data}$$

$$\sim \textbf{Inv.Gamma}\left(\frac{NT + \nu_\sigma}{2}, \frac{(y - D\alpha - X\beta)^\intercal (y - D\alpha - X\beta) + \lambda_\sigma}{2}\right),$$

$$\alpha_0|\boldsymbol{\theta}_{-\alpha_0}, \text{data} \sim \textbf{Normal}\left(\frac{\tau^{-2}\sum_{i=1}^N \alpha_i + \xi^{-2}\mu}{\tau^{-2}N + \xi^{-2}}, \frac{1}{\tau^{-2}N + \xi^{-2}}\right),$$

$$\boldsymbol{\tau}^2|\boldsymbol{\theta}_{-\tau^2}, \text{data}$$

$$\sim \textbf{Inv.Gamma}\left(\frac{N + \nu_\tau}{2}, \frac{(\alpha - \alpha_0\iota)^\intercal (\alpha - \alpha_0\iota) + \lambda_\tau}{2}\right).$$

Non-Centered Parametrization

Let $\tilde{\alpha}_i \triangleq \alpha_i - \alpha_0$ and rewrite the regression model as

$$y_{it} = \alpha_0 + \tilde{\alpha}_i + x_{it}^\intercal \beta + u_{it}.$$

Then the likelihood is rearranged as

$$p(y|X,\theta) \propto (\sigma^2)^{-\frac{NT}{2}} \mathbf{exp}\left[-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\sum_{i=1}^{N}(\tilde{e}_{it} - \alpha_0)^2\right],$$

where $\tilde{e}_{it} = y_{it} - \tilde{\alpha}_i - x_{it}^\intercal \beta$. The hierarchical prior of $\tilde{\alpha}_i$ is

$$\tilde{\alpha}_i \sim \text{Normal}(0, \tau^2).$$

ASIS i

The above model specification is called an ancillary augmentation (AA) while the previous one is called a sufficient augmentation (SA). Both produce the identical posterior distribution, but the sampling efficiency may differ.

We employ an ancillarity-sufficiency interweaving strategy (ASIS) to achieve much higher efficiency of the Gibbs sampler. In ASIS, we first set up two types of Gibbs sampler: one is based on SA and the other is based on AA, and generate the parameters by interweaving the SA-based sampler and the AA-based sampler together.

ASIS ii

For the full conditionals of $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}^2$ in AA, we replace $\boldsymbol{D\alpha}$ with $\boldsymbol{\alpha_0 \iota + D\tilde{\alpha}}$. The full conditionals of $\boldsymbol{\alpha}_0$ and $\boldsymbol{\tau}^2$ in AA are given as follows.

$$\begin{aligned}
&\boldsymbol{\alpha}_0 | \boldsymbol{\theta}_{-\alpha_0}, \text{data} \\
&\qquad \sim \text{Normal}\left(\frac{\sigma^{-2} \sum_{t=1}^{T} \sum_{i=1}^{N} \tilde{e}_{it} + \xi^{-2}\mu}{\sigma^{-2} NT + \xi^{-2}}, \frac{1}{\sigma^{-2} NT + \xi^{-2}} \right), \\
&\boldsymbol{\tau}^2 | \boldsymbol{\theta}_{-\tau^2}, \text{data} \sim \text{Inv.Gamma}\left(\frac{N + \nu_\tau}{2}, \frac{\tilde{\alpha}^\intercal \tilde{\alpha} + \lambda_\tau}{2} \right), \\
&\tilde{\alpha} \triangleq [\tilde{\alpha}_1; \cdots; \tilde{\alpha}_N].
\end{aligned}$$

ASIS iii

ASIS for the panel data regression model

Step 1. Generate $\boldsymbol{\alpha}^{(r+0.5)}$, $\boldsymbol{\beta}^{(r+0.5)}$, $\boldsymbol{\sigma}^{2(r+0.5)}$, $\alpha_0^{(r+0.5)}$ and $\boldsymbol{\tau}^{2(r+0.5)}$ with the SA-based sampler.

Step 2. Compute $\tilde{\alpha}_i^{(r+0.5)} = \alpha_i^{(r+0.5)} - \alpha_0^{(r+0.5)}$.

Step 3. Generate $\boldsymbol{\beta}^{(r+1)}$, $\boldsymbol{\sigma}^{2(r+1)}$, $\alpha_0^{(r+1)}$ and $\boldsymbol{\tau}^{2(r+1)}$ with the AA-based sampler.

Step 4. Compute $\alpha_i^{(r+1)} = \tilde{\alpha}_i^{(r+0.5)} + \alpha_0^{(r+1)}$.

Metropolis–Hastings Algorithm

Another popular Markov chain sampling scheme is the Metropolis-Hastings algorithm or MH algorithm in short.

Suppose we want to generate a random variate $\boldsymbol{\theta}$, which may be either scalar or vector, with the target density $f(\boldsymbol{\theta})$, but there exists no easy way to do so.

In Bayesian analysis, $f(\boldsymbol{\theta})$ is the posterior density.

Further suppose there exits a Markov chain with kernel $q(\boldsymbol{\varphi}, \boldsymbol{\theta})$ from which $\boldsymbol{\theta}$ can be easily generated.

Then the MH algorithm is defined as follows.

Metropolis-Hastings Algorithm

Step 1. Set the initial value $\boldsymbol{\theta}^{(0)}$ and let $r = 1$.

Step 2. Draw $\tilde{\boldsymbol{\theta}}$, a candidate for $\boldsymbol{\theta}^{(r)}$, from $q(\boldsymbol{\theta}^{(r-1)}, \boldsymbol{\theta})$.

Step 3. Update $\boldsymbol{\theta}^{(r)}$ with

$$\tilde{u} \leftarrow \mathbf{Uniform}(0,1),$$

$$\boldsymbol{\theta}^{(r)} = \begin{cases} \tilde{\boldsymbol{\theta}}, & (\tilde{u} \leqq \alpha(\boldsymbol{\theta}^{(r-1)}, \tilde{\boldsymbol{\theta}})); \\ \boldsymbol{\theta}^{(r-1)}, & (\tilde{u} > \alpha(\boldsymbol{\theta}^{(r-1)}, \tilde{\boldsymbol{\theta}})), \end{cases}$$

where

$$\alpha(\boldsymbol{\theta}^{(r-1)}, \tilde{\boldsymbol{\theta}}) = \min\left\{ \frac{f(\tilde{\boldsymbol{\theta}}) q(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta}^{(r-1)})}{f(\boldsymbol{\theta}^{(r-1)}) q(\boldsymbol{\theta}^{(r-1)}, \tilde{\boldsymbol{\theta}})}, 1 \right\}.$$

Step 4. Increase r by 1 and go to **Step 2.**

Advantages Of MH Algorithms i

1. Replacing $f(\theta)$ with the posterior distribution $p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$, we have

$$\alpha(\theta^{(r-1)}, \tilde{\theta}) = \min\left\{ \frac{\frac{p(D|\tilde{\theta})p(\tilde{\theta})}{p(D)} q(\tilde{\theta}, \theta^{(r-1)})}{\frac{p(D|\theta^{(r-1)})p(\theta^{(r-1)})}{p(D)} q(\theta^{(r-1)}, \tilde{\theta})}, 1 \right\}$$
$$= \min\left\{ \frac{p(D|\tilde{\theta})p(\tilde{\theta})q(\tilde{\theta}, \theta^{(r-1)})}{p(D|\theta^{(r-1)})p(\theta^{(r-1)})q(\theta^{(r-1)}, \tilde{\theta})}, 1 \right\}.$$

Thus we do not need to know the exact expression of the marginal likelihood $p(D)$ in applying the MH algorithm.

Advantages Of MH Algorithms ii

2. A sufficient condition for the MH algorithm to work is that $q(\varphi, \theta) > 0$ for any $(\varphi, \theta) \in \Theta \times \Theta$ where Θ is the support of f, i.e., $\Theta = \{\theta \in \mathbb{R}^k : f(\theta) > 0\}$. This is indeed sufficient in most applications.
3. Instead of applying the MH step to θ at once, we may apply it element-by-element or block-by-block.
4. Furthermore, if the full conditional of a certain element or block of θ is available, we may draw it directly from its full conditional. In this sense, the Gibbs sampler is a special case of the MH algorithm.

Derivation Of The Invariant Density i

For simplicity, we denote $\boldsymbol{\varphi} = \boldsymbol{\theta}^{(r-1)}$ and $\boldsymbol{\theta} = \boldsymbol{\theta}^{(r)}$. First, we derive the transition kernel of the MH algorithm. If $\tilde{\boldsymbol{\theta}}$ is accepted, the probability of $\tilde{\boldsymbol{\theta}} \in A \subseteqq \boldsymbol{\Theta}$ is given by

$$
\begin{aligned}
&\mathbf{P}(\tilde{\theta} \in A|\text{Accepted}) \\
&= \mathbf{P}(\tilde{\theta} \in A|\tilde{u} \leqq \alpha(\varphi, \tilde{\theta})) = \frac{\mathbf{P}(\tilde{\theta} \in A, \tilde{u} \leqq \alpha(\varphi, \tilde{\theta}))}{\mathbf{P}(\tilde{u} \leqq \alpha(\varphi, \tilde{\theta}))} \\
&= \frac{\int_A \int_0^{\alpha(\varphi,\theta)} q(\varphi, \theta) du d\theta}{\int_{\Theta} \int_0^{\alpha(\varphi,\theta)} q(\varphi, \theta) du d\theta} = \frac{\int_A \alpha(\varphi, \theta) q(\varphi, \theta) d\theta}{\int_{\Theta} \alpha(\varphi, \theta) q(\varphi, \theta) d\theta} \\
&= \int_A \frac{\alpha(\varphi, \theta) q(\varphi, \theta)}{\alpha(\varphi)} d\theta, \quad \alpha(\varphi) = \int_{\Theta} \alpha(\varphi, \theta) q(\varphi, \theta) d\theta.
\end{aligned}
$$

$\alpha(\varphi)$ is the "average" probability that $\tilde{\theta}$ is accepted in Step 3. of the MH algorithm. Therefore the conditional p.d.f. of θ given that it is accepted is

$$f(\theta|\varphi, \text{Accepted}) = \frac{\alpha(\varphi, \theta)q(\varphi, \theta)}{\alpha(\varphi)}.$$

The conditional p.d.f. of θ given that it is rejected is

$$f(\theta|\varphi, \text{Rejected}) = \delta(||\theta - \varphi||),$$

where $\delta(\cdot)$ is the Dirac delta function and $||\cdot||$ is the Euclidean norm.

Derivation Of The Invariant Density iii

Since the conditional p.d.f. of $\boldsymbol{\theta}$ given $\boldsymbol{\varphi}$ is expressed a mixture of two conditional p.d.f.'s:

$$\begin{aligned} f(\boldsymbol{\theta}|\boldsymbol{\varphi}) = \alpha(\boldsymbol{\varphi}) f(\boldsymbol{\theta}|\boldsymbol{\varphi}, \text{Accepted}) \\ + \{1-\alpha(\boldsymbol{\varphi})\} f(\boldsymbol{\theta}|\boldsymbol{\varphi}, \text{Rejected}), \end{aligned}$$

the transition kernel of the MH algorithm is derived as

$$K(\boldsymbol{\varphi}, \boldsymbol{\theta}) = \alpha(\boldsymbol{\varphi}, \boldsymbol{\theta}) q(\boldsymbol{\varphi}, \boldsymbol{\theta}) + \{1-\alpha(\boldsymbol{\varphi})\}\delta(\|\boldsymbol{\theta}-\boldsymbol{\varphi}\|).$$

Derivation Of The Invariant Density iv

As the final step of the proof, it is sufficient to show that the transition kernel $K(\varphi, \theta)$ and the target density $f(\theta)$ satisfy the detailed balance condition:

$$f(\varphi)K(\varphi, \theta) = f(\theta)K(\theta, \varphi).$$

By multiplying both sides of the detailed balance condition with $f(\varphi)$ from the left, we have

$$\begin{aligned} f(\varphi)K(\varphi, \theta) = \alpha(\varphi, \theta)f(\varphi)q(\varphi, \theta) \\ + \{1 - \alpha(\varphi)\}f(\varphi)\delta(\|\theta - \varphi\|). \end{aligned}$$

The first term in the right-hand side is

$$\begin{aligned}
\alpha(\varphi,\theta)f(\varphi)q(\varphi,\theta) &= \min\left\{\frac{f(\theta)q(\theta,\varphi)}{f(\varphi)q(\varphi,\theta)}, 1\right\} f(\varphi)q(\varphi,\theta) \\
&= \min\{f(\theta)q(\theta,\varphi), f(\varphi)q(\varphi,\theta)\} \\
&= \min\left\{\frac{f(\varphi)q(\varphi,\theta)}{f(\theta)q(\theta,\varphi)}, 1\right\} f(\theta)q(\theta,\varphi) \\
&= \alpha(\theta,\varphi)f(\theta)q(\theta,\varphi).
\end{aligned}$$

The second term in the right-hand side is

$$\{1-\alpha(\varphi)\}f(\varphi)\delta(||\theta-\varphi||) = \{1-\alpha(\theta)\}f(\theta)\delta(||\varphi-\theta||).$$

Thus the detailed balance condition is satisfied. ■

Random Walk Chain i

Although we may pick virtually any $q(\varphi, \theta)$ in theory, the choice of $q(\varphi, \theta)$ is crucial to the efficiency of the MH algorithm.

One of the popular choices for $q(\varphi, \theta)$ is a random walk process:

$$\theta = \varphi + \epsilon, \quad \mathrm{E}[\epsilon] = 0, \quad \mathrm{Var}[\epsilon] = \Sigma,$$

where ϵ is a white noise. In many applications, we suppose the transition kernel is symmetric, i.e.,

$$q(\varphi, \theta) = q(\theta, \varphi).$$

Random Walk Chain ii

For example, in case of $\boldsymbol{\epsilon} \sim \mathbf{Normal}(\mathbf{0}, \boldsymbol{\Sigma})$, this symmetry holds since

$$q(\varphi, \theta) = (2\pi)^{-\frac{k}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}(\theta - \varphi)^\intercal \boldsymbol{\Sigma}^{-1}(\theta - \varphi)\right],$$

Then the acceptance probability becomes

$$\alpha(\varphi, \theta) = \min\left\{\frac{f(\theta)}{f(\varphi)}, 1\right\}.$$

Independent Chain i

Another popular choice for $q(\varphi, \theta)$ is a sequence of i.i.d. random variables, i.e.,

$$q(\varphi, \theta) = g(\theta),$$

where g is a p.d.f. of θ and it does not depend on φ.

Such g is called the proposal density.

Since a sequence of i.i.d. random variables is a special case of Markov chain, we apply the same argument to this case as a more general Markov chain $q(\varphi, \theta)$.

Independent Chain ii

Then the acceptance probability is

$$\alpha(\varphi, \theta) = \min\left\{\frac{f(\theta)g(\varphi)}{f(\varphi)g(\theta)}, 1\right\}.$$

This type of MH algorithm is called the independent chain, though it is not an independent process because the acceptance probability $\alpha(\varphi, \theta)$ depends on the past realization of the parameter φ.

If f and g are identical, the acceptance probability is identical to 1. Thus in practice we try to make g as close to f as possible.

Independent Chain iii

One widely applied technique to approximate f is to apply the second-order Taylor approximation to $\log f(\theta)$:

$$\begin{aligned}\log f(\theta) \approx \log f(\theta^*) &+ \nabla_\theta \log f(\theta^*)^\intercal (\theta - \theta^*) \\ &+ \frac{1}{2}(\theta - \theta^*)^\intercal \nabla_\theta^2 \log f(\theta^*)(\theta - \theta^*),\end{aligned}$$

where θ^* is the mode of the target distribution. In case $f(\theta)$ is the posterior density, θ^* is the MAP estimator.

Note that the first-order condition is

$$\nabla_\theta \log f(\theta^*) = 0.$$

Then we have

$$f(\theta) \approx \mathcal{K}^* (2\pi)^{-\frac{k}{2}} |\Sigma^*|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}(\theta - \theta^*)^\intercal \left(\Sigma^*\right)^{-1} (\theta - \theta^*)\right],$$

$$\Sigma^* \triangleq -\left(\nabla_\theta^2 \log f(\theta^*)\right)^{-1}, \quad \mathcal{K}^* \triangleq f(\theta^*)(2\pi)^{\frac{k}{2}} |\Sigma^*|^{\frac{1}{2}}.$$

Therefore, if the second-order Taylor approximation is good enough, we may use the following normal distribution

$$\theta \sim \textbf{Normal}\left(\theta^*, \Sigma^*\right),$$

as the proposal density g.

Example: AR(1) Process i

Consider an AR(1) process:

$$y_t = \phi y_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathbf{Normal}(0, \sigma^2),$$

where we suppose $|\phi| < 1$ so that the AR(1) process is stationary. Then the invariant distribution is

$$y_1 \sim \mathbf{Normal}\left(0, \frac{\sigma^2}{1-\phi^2}\right).$$

Example: AR(1) Process ii

Thus the joint distribution of $D = \{y_1, \ldots, y_n\}$ or the likelihood of (ϕ, σ^2) given D is

$$p(D|\phi,\sigma^2) = \sqrt{\frac{1-\phi^2}{2\pi\sigma^2}}\exp\left[-\frac{(1-\phi^2)y_1^2}{2\sigma^2}\right]$$
$$\times \prod_{t=2}^{n} \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(y_t-\phi y_{t-1})^2}{2\sigma^2}\right].$$

The likelihood is rearranged as

$$
\begin{aligned}
&p(D|\phi,\sigma^2)\\
&\propto \sqrt{1-\phi^2}\sigma^{-\frac{n}{2}} \exp\left[-\frac{(1-\phi^2)y_1^2 + \sum_{t=2}^{n}(y_t - \phi y_{t-1})^2}{2\sigma^2}\right]\\
&\propto \sqrt{1-\phi^2}\sigma^{-\frac{n}{2}} \exp\left[-\frac{\sum_{t=1}^{n} y_t^2 - 2\phi \sum_{t=2}^{n} y_t y_{t-1} + \phi^2 \sum_{t=2}^{n-1} y_t^2}{2\sigma^2}\right].
\end{aligned}
$$

The prior is

$$
\frac{\phi+1}{2} \sim \mathrm{Beta}(\alpha_\phi, \beta_\phi), \quad \sigma^2 \sim \mathrm{Inv.Gamma}\left(\frac{\alpha_\sigma}{2}, \frac{\beta_\sigma}{2}\right).
$$

Example: AR(1) Process iv

The full conditional of $\boldsymbol{\sigma}^2$ is

$$\sigma^2|\phi, D \sim \textbf{Inv.Gamma}\Bigg(\frac{n+\alpha_\sigma}{2},$$
$$\frac{1}{2}\left\{\sum_{t=1}^{n} y_t^2 - 2\phi\sum_{t=2}^{n} y_t y_{t-1} + \phi^2\sum_{t=2}^{n-1} y_t^2 + \beta_\sigma\right\}\Bigg),$$

while the full conditional of $\boldsymbol{\phi}$ is

$$p(\phi|\sigma^2, D) \propto \sqrt{1-\phi^2}(1+\phi)^{\alpha_\phi - 1}(1-\phi)^{\beta_\phi - 1}$$
$$\times \exp\left[-\frac{\sum_{t=2}^{n-1} y_t^2}{2\sigma^2}\left(\phi - \frac{\sum_{t=2}^{n} y_t y_{t-1}}{\sum_{t=2}^{n-1} y_t^2}\right)^2\right]\mathbb{1}_{(-1,1)}(\phi).$$

Example: AR(1) Process v

Although the full conditional of $\boldsymbol{\phi}$ is not standard, if we ignore $\sqrt{1-\boldsymbol{\phi}^2}(1+\boldsymbol{\phi})^{\alpha_\phi-1}(1-\boldsymbol{\phi})^{\beta_\phi-1}$, it is proportional to the p.d.f. of the truncated normal distribution:

$$\boldsymbol{\phi}|\boldsymbol{\sigma}^2, D \sim \mathbf{Normal}_{(-1,1)}\left(\frac{\sum_{t=2}^{n} y_t y_{t-1}}{\sum_{t=2}^{n-1} y_t^2}, \frac{\boldsymbol{\sigma}^2}{\sum_{t=2}^{n-1} y_t^2}\right).$$

Thus we may use the above normal distribution as the proposal distribution for $\boldsymbol{\phi}$ in the MH step.

In particular, when $\alpha_\phi = \beta_\phi = \frac{1}{2}$, we have

$$\begin{aligned}\sqrt{1-\phi^2}&(1+\phi)^{\alpha_\phi - 1}(1-\phi)^{\beta_\phi - 1}\\ &= (1+\phi)^{\frac{1}{2}}(1-\phi)^{\frac{1}{2}}(1+\phi)^{-\frac{1}{2}}(1-\phi)^{-\frac{1}{2}} = 1.\end{aligned}$$

Thus the full conditional of ϕ is exactly identical to the above truncated normal distribution.

Note that the beta prior with $\alpha_\phi = \beta_\phi = \frac{1}{2}$ is the Jeffreys prior.

Acceptance-Rejection Method

Suppose we cannot generate a random variable $\boldsymbol{\theta}$ directly from $f(\boldsymbol{\theta})$ but it is possible to do so from $g(\boldsymbol{\theta})$ such that $f(\boldsymbol{\theta}) \leqq \mathcal{K}g(\boldsymbol{\theta})$ for any $\boldsymbol{\theta} \in \boldsymbol{\Theta}$.

Acceptance-Rejection Method

Step 1. Generate $\tilde{\boldsymbol{\theta}}$ from $g(\boldsymbol{\theta})$.

Step 2. Generate $\tilde{u}$ from $\mathbf{Uniform}(0,1)$.

Step 3. Accept $\tilde{\boldsymbol{\theta}}$ if

$$\tilde{u} \leqq \frac{f(\tilde{\boldsymbol{\theta}})}{\mathcal{K}g(\tilde{\boldsymbol{\theta}})};$$

otherwise, go to Step 1.

Proof Of The Acceptance-Rejection Method i

For simplicity, we prove it when $\boldsymbol{\theta}$ is a scaler.

Let $\tilde{F}(\boldsymbol{\theta})$ denote the c.d.f. of $\tilde{\boldsymbol{\theta}}$ generated with the acceptance-rejection method. Then

$$\tilde{F}(\theta) = \mathrm{P}\left(\tilde{\theta} \leqq \theta \,\middle|\, \tilde{u} \leqq \frac{f(\tilde{\theta})}{\mathcal{K}g(\tilde{\theta})}\right) = \frac{\mathrm{P}\left(\tilde{\theta} \leqq \theta, \tilde{u} \leqq \frac{f(\tilde{\theta})}{\mathcal{K}g(\tilde{\theta})}\right)}{\mathrm{P}\left(\tilde{u} \leqq \frac{f(\tilde{\theta})}{\mathcal{K}g(\tilde{\theta})}\right)}.$$

Proof Of The Acceptance-Rejection Method ii

Note that

$$
\begin{aligned}
\mathbf{P}\left(\tilde{\theta} \leqq \theta, \tilde{u} \leqq \frac{f(\tilde{\theta})}{\mathcal{K} g(\tilde{\theta})}\right) &= \int_{-\infty}^{\theta} \int_{0}^{f(y)/(\mathcal{K} g(y))} g(y) du dy \\
&= \int_{-\infty}^{\theta} \frac{f(y)}{\mathcal{K} g(y)} g(y) dy \\
&= \mathcal{K}^{-1} \int_{-\infty}^{\theta} f(y) dy \\
&= \mathcal{K}^{-1} F(\theta),
\end{aligned}
$$

and

$$\begin{aligned}\mathbf{P}\left(\tilde{u} \leqq \frac{f(\tilde{\theta})}{\mathcal{K}g(\tilde{\theta})}\right) &= \int_{-\infty}^{\infty} \int_{0}^{f(y)/(\mathcal{K}g(y))} g(y)dudy \\ &= \int_{-\infty}^{\infty} \frac{f(y)}{\mathcal{K}g(y)} g(y)dy \\ &= \mathcal{K}^{-1},\end{aligned}$$

Therefore

$$\tilde{F}(\theta) = \frac{\mathcal{K}^{-1}F(\theta)}{\mathcal{K}^{-1}} = F(\theta).$$

■

Rejection Sampling Chain

Unfortunately, such $g(\theta)$ does not necessarily exist in practice. Instead, we my apply the following algorithm.

Rejection Sampling Chain

Step 1. Generate $\tilde{\theta}$ with the acceptance-rejection method.

Step 2. Generate $\tilde{u}$ from $\mathbf{Uniform}(0, 1)$.

Step 3. Accept $\tilde{\theta}$ as $\theta^{(r)}$ from f with probability:

$$\alpha(\theta^{(r-1)}, \tilde{\theta}) = \begin{cases} \min\left\{\frac{Kg(\theta^{(r-1)})}{f(\theta^{(r-1)})}, 1\right\} & \text{if } f(\tilde{\theta}) \leqq Kg(\tilde{\theta}); \\ \min\left\{\frac{f(\tilde{\theta})g(\theta^{(r-1)})}{f(\theta^{(r-1)})g(\tilde{\theta})}, 1\right\} & \text{if } f(\tilde{\theta}) > Kg(\tilde{\theta}). \end{cases}$$

Let $\varphi = \theta^{(r-1)}$ and $\theta = \tilde{\theta}$. Apply the acceptance-rejection method though $f(\theta) \leqq Kg(\theta)$ does not always hold. Then the resultant random variable has the following p.d.f. :

$$\tilde{f}(\theta) \propto \min\{f(\theta), Kg(\theta)\}.$$

Thus the acceptance probability in the MH step becomes

$$\begin{aligned}\alpha(\varphi, \theta) &= \min\left\{\frac{f(\theta)\min\{f(\varphi), Kg(\varphi)\}}{f(\varphi)\min\{f(\theta), Kg(\theta)\}}, 1\right\} \\ &= \min\left\{\frac{\min\{Kg(\varphi)/f(\varphi), 1\}}{\min\{Kg(\theta)/f(\theta), 1\}}, 1\right\}.\end{aligned}$$

Proof Of The Rejection Sampling Chain ii

If $f(\theta) \leqq Kg(\theta)$, $\min\{Kg(\theta)/f(\theta), 1\} = 1$. Therefore

$$\alpha(\varphi, \theta) = \min\left\{\frac{Kg(\varphi)}{f(\varphi)}, 1\right\}.$$

If $f(\theta) > Kg(\theta)$, $\min\{Kg(\theta)/f(\theta), 1\} = Kg(\theta)/f(\theta)$. Hence

$$\begin{aligned}\alpha(\varphi, \theta) &= \min\left\{\min\left\{\frac{Kg(\varphi)}{f(\varphi)}\frac{f(\theta)}{Kg(\theta)}, \frac{f(\theta)}{Kg(\theta)}\right\}, 1\right\}\\ &= \min\left\{\frac{f(\theta)g(\varphi)}{f(\varphi)g(\theta)}, 1\right\}.\end{aligned}$$

$\blacksquare$

Example: Taylor Approximation

When we apply the Taylor-approximation to f, we have the following relationship between f and g:

$$f(\theta) \approx \mathcal{K}^* g(\theta),$$

$$g(\theta) = (2\pi)^{-\frac{k}{2}} |\Sigma^*|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}(\theta - \theta^*)^\top \left(\Sigma^*\right)^{-1} (\theta - \theta^*)\right],$$

$$\Sigma^* \triangleq -\left(\nabla_\theta^2 \log f(\theta^*)\right)^{-1}, \quad \mathcal{K}^* \triangleq f(\theta^*)(2\pi)^{\frac{k}{2}} |\Sigma^*|^{\frac{1}{2}}.$$

In this case, we may apply the rejection sampling chain with $\mathcal{K}^*$. The acceptance-rejection method is applicable for the region $\{\theta : f(\theta) \leqq \mathcal{K}^* g(\theta)\}$. Therefore some improvement in the acceptance probability would be expected if the approximation is good enough.

Example: Gamma Distribution i

Suppose $y_1, \ldots, y_n$ are independently drawn from $\mathbf{Gamma}(\alpha, \beta)$. The likelihood is

$$p(D|\theta) = \prod_{i=1}^{n} \frac{\beta^\alpha}{\Gamma(\alpha)} y_i^{\alpha-1} e^{-\beta y_i}.$$

To make the sampling procedure more stable, we change the parameterization from (α, β) to $(\alpha, \alpha/\gamma)$. Then

$$p(D|\theta) = \frac{\alpha^{\alpha n}\gamma^{-\alpha n}}{\Gamma(\alpha)^n} \exp\left[(\alpha - 1)\sum_{i=1}^{n} \log y_i - \frac{\alpha}{\gamma}\sum_{i=1}^{n} y_i\right].$$

Example: Gamma Distribution ii

We assume the following priors:

$$\alpha \sim \mathbf{Gamma}(a_\alpha, b_\alpha), \quad \gamma \sim \mathbf{Inv.Gamma}(a_\gamma, b_\gamma).$$

The full conditional of γ is derived as a gamma distribution:

$$\gamma | \alpha, D \sim \mathbf{Inv.Gamma}\left(\alpha n + a_\gamma, \alpha \sum_{i=1}^{n} y_i + b_\gamma\right).$$

The full conditional of α, on the other hand, is non-standard one:

$$p(\alpha|\gamma, D) \propto \underbrace{\frac{\alpha^{\alpha n}}{\Gamma(\alpha)^n} \alpha^{a_\alpha - 1} e^{-\alpha(\eta + b_\alpha)}}_{f(\alpha)}, \; \eta \triangleq \sum_{i=1}^{n} \left(\frac{y_i}{\gamma} - \log \frac{y_i}{\gamma}\right).$$

Example: Gamma Distribution iii

Define

$$\begin{aligned}
\log f(\alpha) &= n(\alpha \log \alpha - \log \Gamma(\alpha)) \\
&\quad + (a_\alpha - 1)\log \alpha - \alpha(\eta + b_\alpha), \\
\nabla \log f(\alpha) &= n(1 + \log \alpha - \psi^{(0)}(\alpha)) + \frac{a_\alpha - 1}{\alpha} - \eta - b_\alpha, \\
\nabla^2 \log f(\alpha) &= n\left(\frac{1}{\alpha} - \psi^{(1)}(\alpha)\right) - \frac{a_\alpha - 1}{\alpha^2},
\end{aligned}$$

where $\psi^{(s)}$ is the polygamma function of order s.

Remark: $\nabla^2 \log f(\alpha) < 0$ if $n + 2a_\gamma > 2$.

Example: Gamma Distribution iv

Applying the second-order Taylor approximation to $\log f(\alpha)$ around the mode $\alpha^* = \arg\max \log f(\alpha)$, we have

$$\log f(\alpha) \approx \log f(\alpha^*) + \frac{1}{2}\nabla^2 \log f(\alpha^*)(\alpha - \alpha^*)^2.$$

Since $\alpha > 0$, a suitable proposal distribution would be

$$\alpha|\gamma, D \sim \mathbf{Normal}^+\left(\alpha^*, \frac{1}{-\nabla^2 \log f(\alpha^*)}\right),$$

where $\mathbf{Normal}^+(\cdot)$ is the positively truncated normal distribution.

We may apply the rejection sampling chain with the above truncated normal distribution.